

Lecture 1

Unit 1 Time Series

By Sanchit, Rishabh and Shobhit



Join the lecture online on *your* dashboard.

Let's start with a minute of silence.

आचार्यात् पादं आधत्ते पादं शिष्यः स्वमेधया ।
पादं सब्रह्मचारिभ्यः पादं कालक्रमेण च ॥

Meaning:

A student acquires knowledge in four equal parts:

- One-fourth from the teacher
- One-fourth through self-reflection and independent thinking
- One-fourth through discussions with peers and fellow learners
- One-fourth over time through personal experience

Recap

- Why Advance ML?
- What is ML?
- Course Objectives
- Roadmap
- Evaluation and Classroom Rules
- Required Readings and References

Agenda

- What is Time Series?
- Components of Time Series Data
- Decomposition of Time Series Data

City- Energy Demand Forecasting

- **Problem Statement**

- A **power distribution company** must forecast electricity demand at multiple horizons:
 - 15-minute ahead (real-time load balancing)
 - Day-ahead (generation planning)
 - Week-ahead (maintenance scheduling)
- **Wrong forecasts** lead to **blackouts** or **wasted** generation cost.

- **Dataset Link:** https://data.open-power-system-data.org/time_series/

City- Energy Demand Forecasting

Assumed Data & Features

Assume we have 15-minute interval data with:

- Target load → electricity demand (MW)

Time-based features

- hour
- day_of_week
- is_weekend
- month

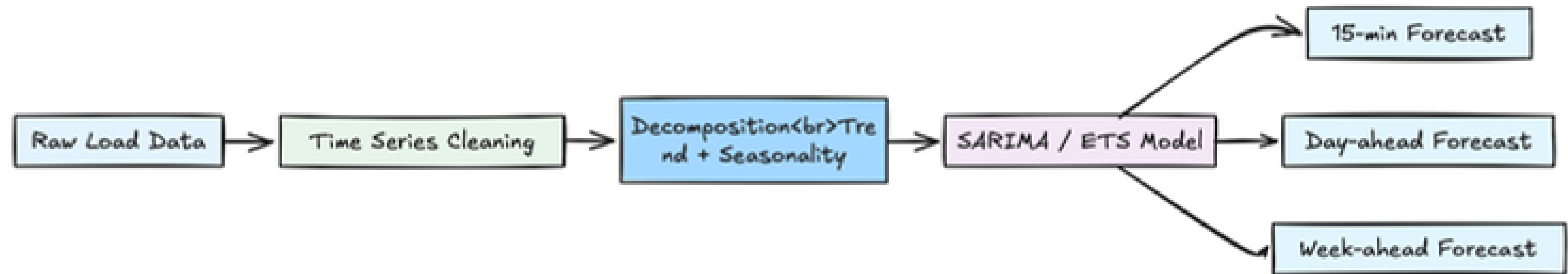
Historical features

load_{t-1} (last 15 min)
load_{t-4} (1 hour ago)
load_{t-96} (same time yesterday)
rolling_mean_1h
rolling_mean_24h

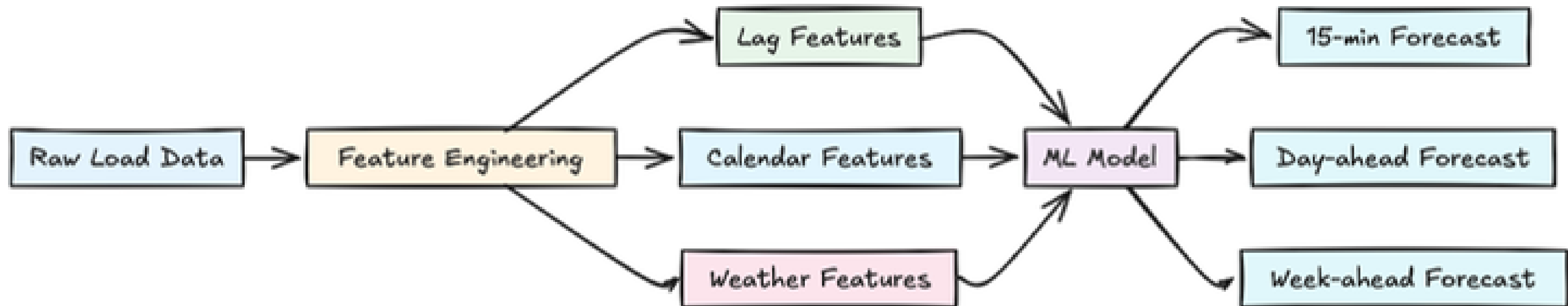
External features (optional but realistic)

temperature
humidity
holiday_flag

City- Energy Demand Forecasting

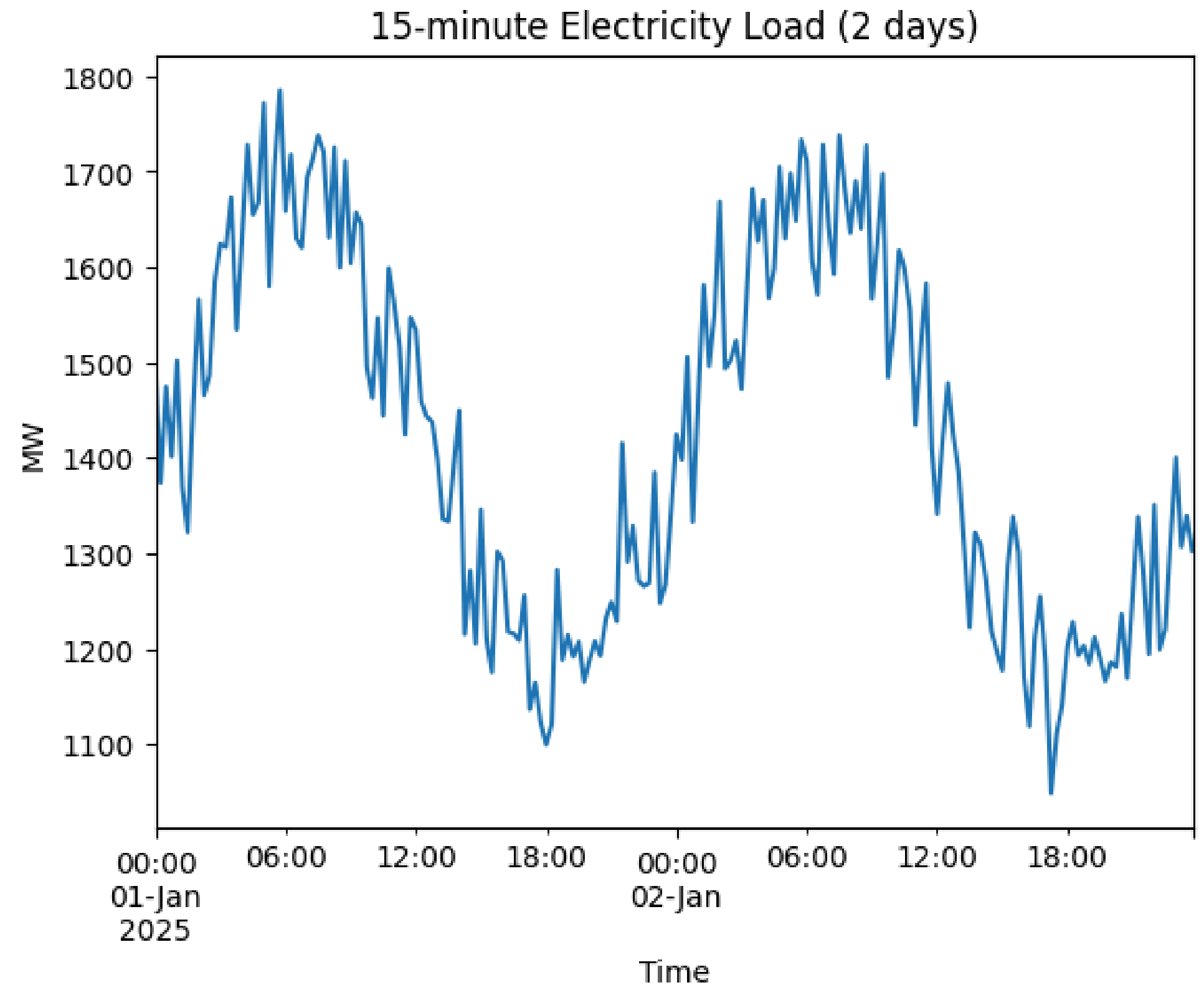


City- Energy Demand Forecasting



City- Energy Demand Forecasting

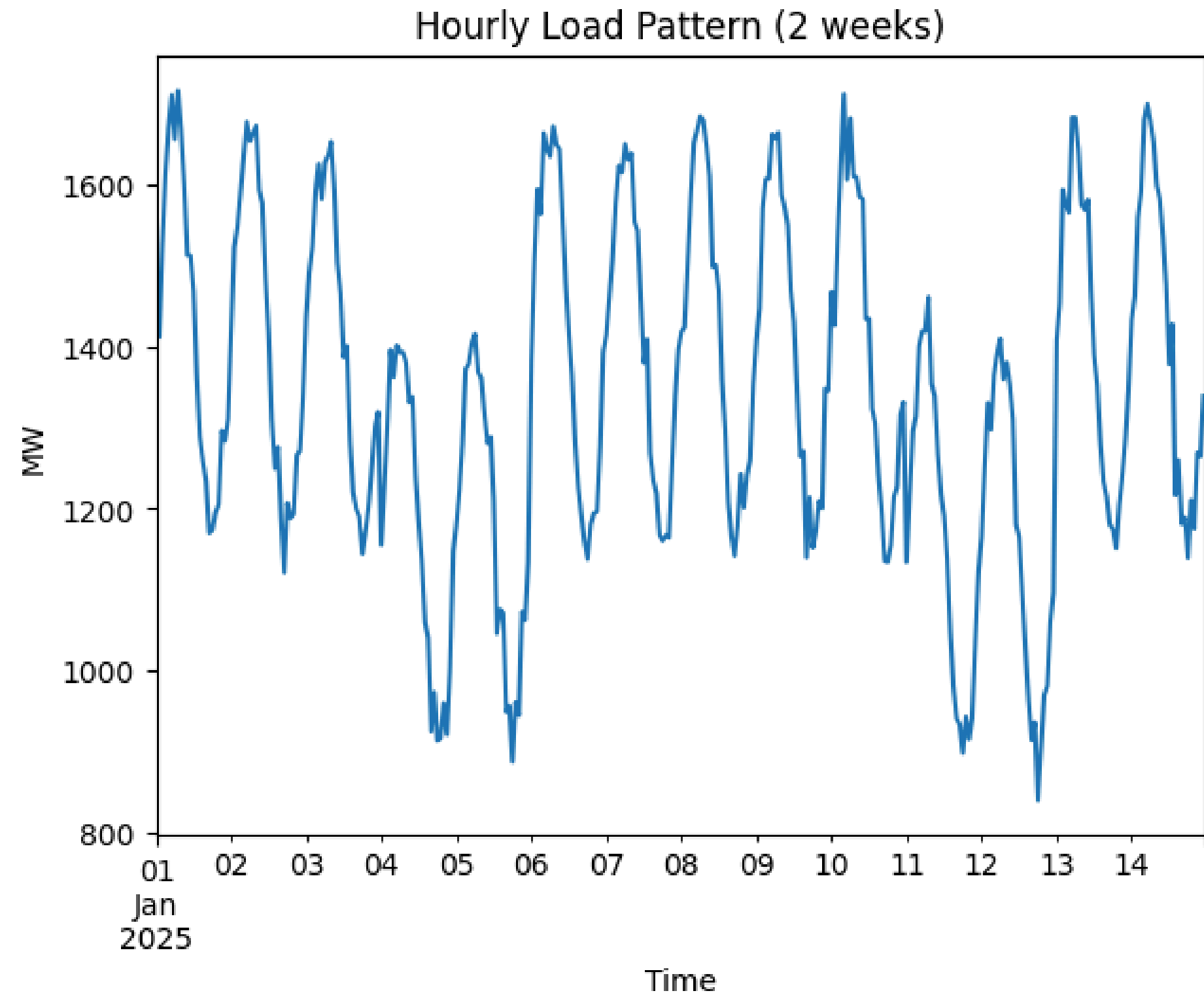
15-minute
Electricity Load for
2 days



What do you notice as you look from left to right in this plot?

City- Energy Demand Forecasting

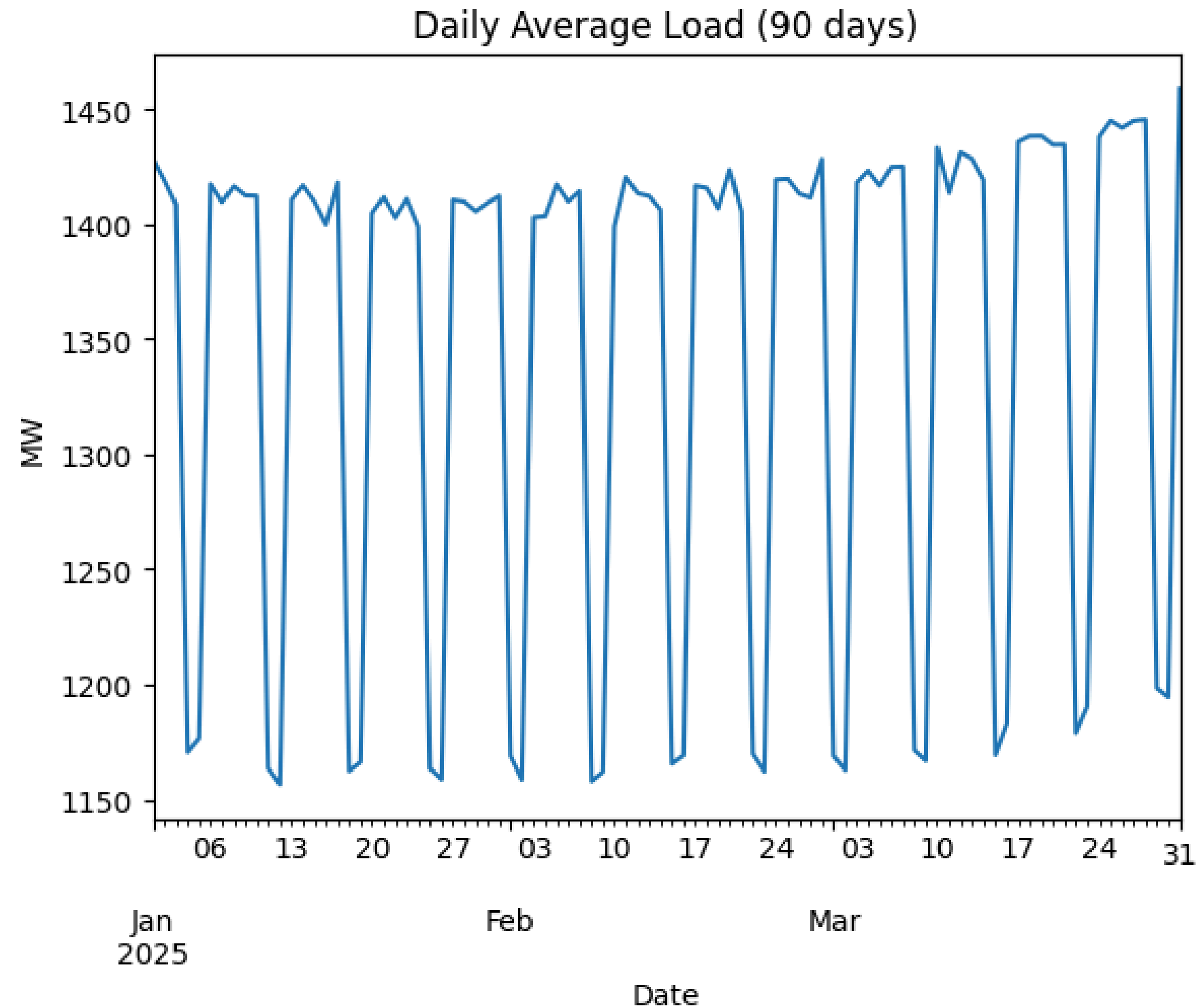
Hourly load for
2 weeks



What do you notice as you look from left to right in this plot?

City- Energy Demand Forecasting

Daily average
for 90 days

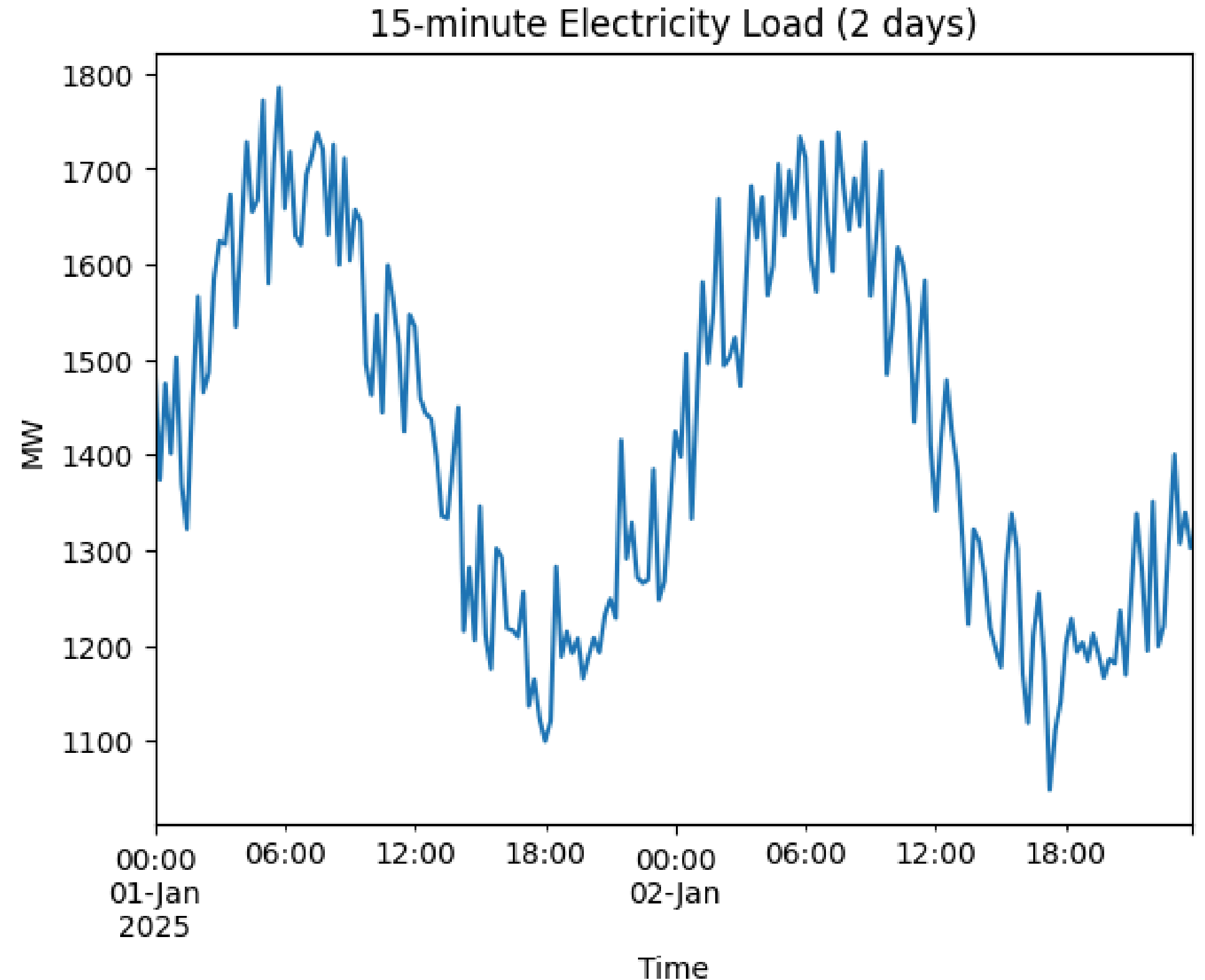


What do you notice as you look from left to right in this plot?

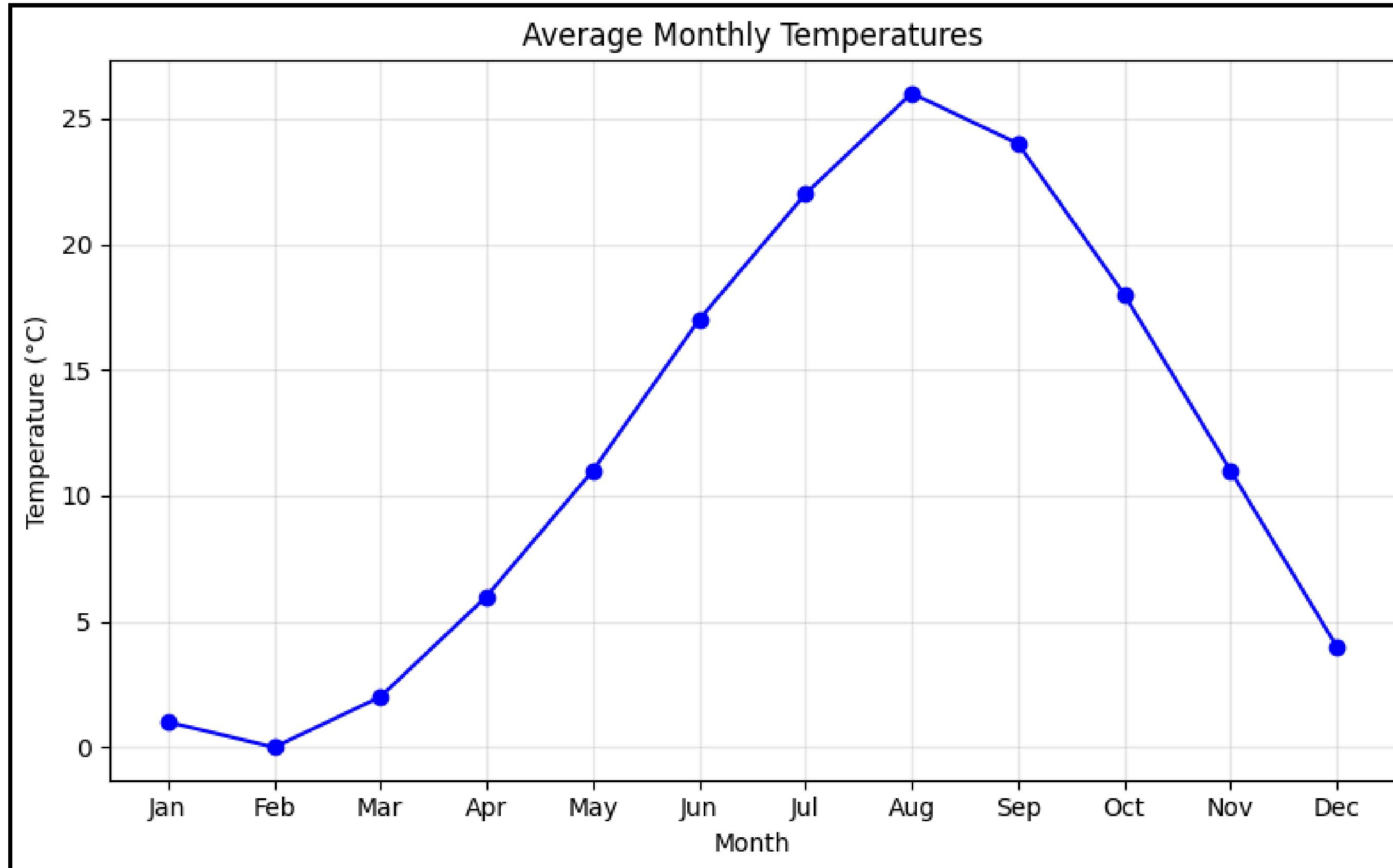
**What is the relation between time and
electricity load/consumption?**

What is the relation between time and electricity load/consumption?

- As time increases, there is a pattern in the consumption.
- The consumption is highest at 6:00 and lowest at 18:00.
- 1 Jan 2025 consumption is very similar to 2nd jan 2025

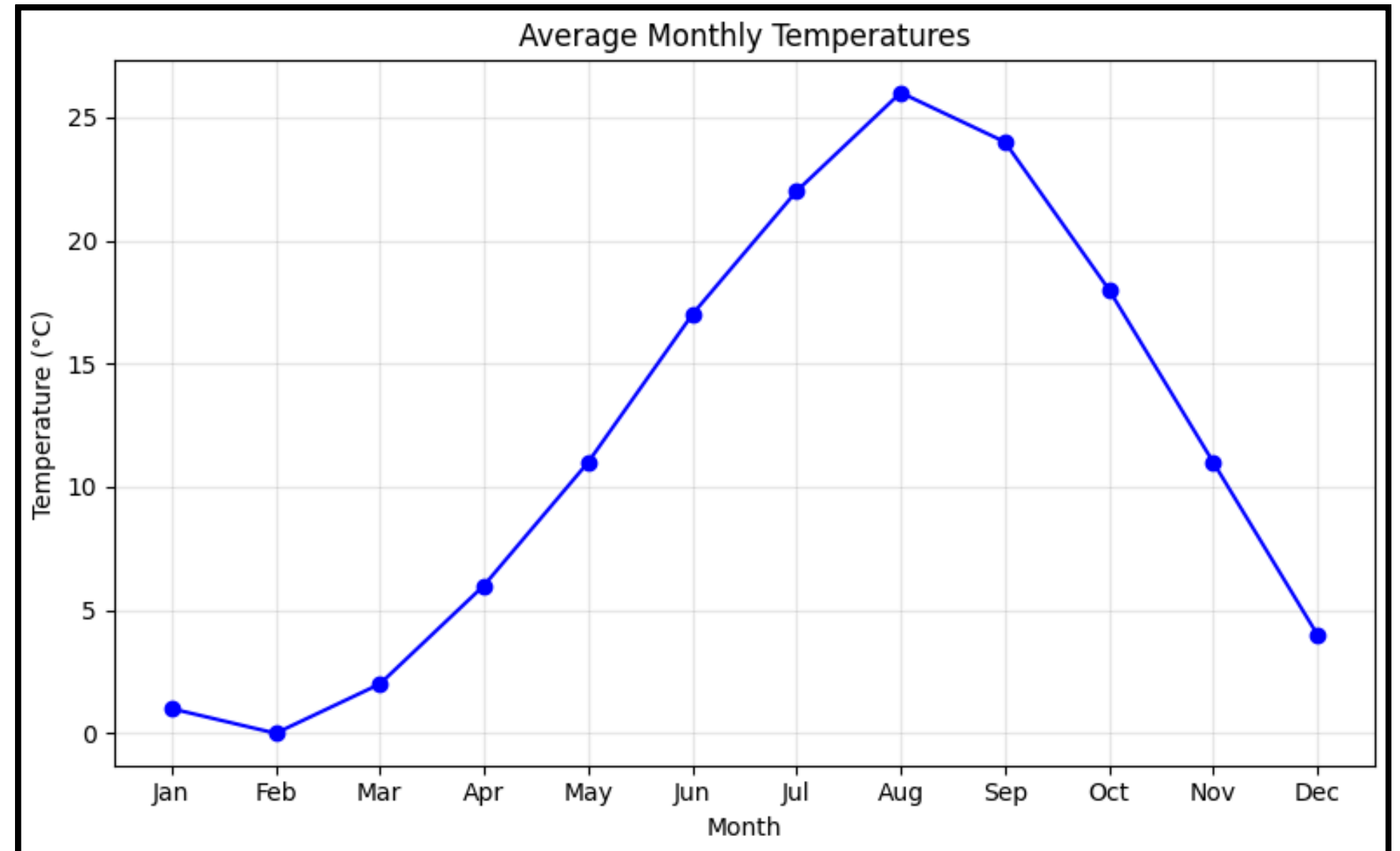


What about this graph between time and temperature?



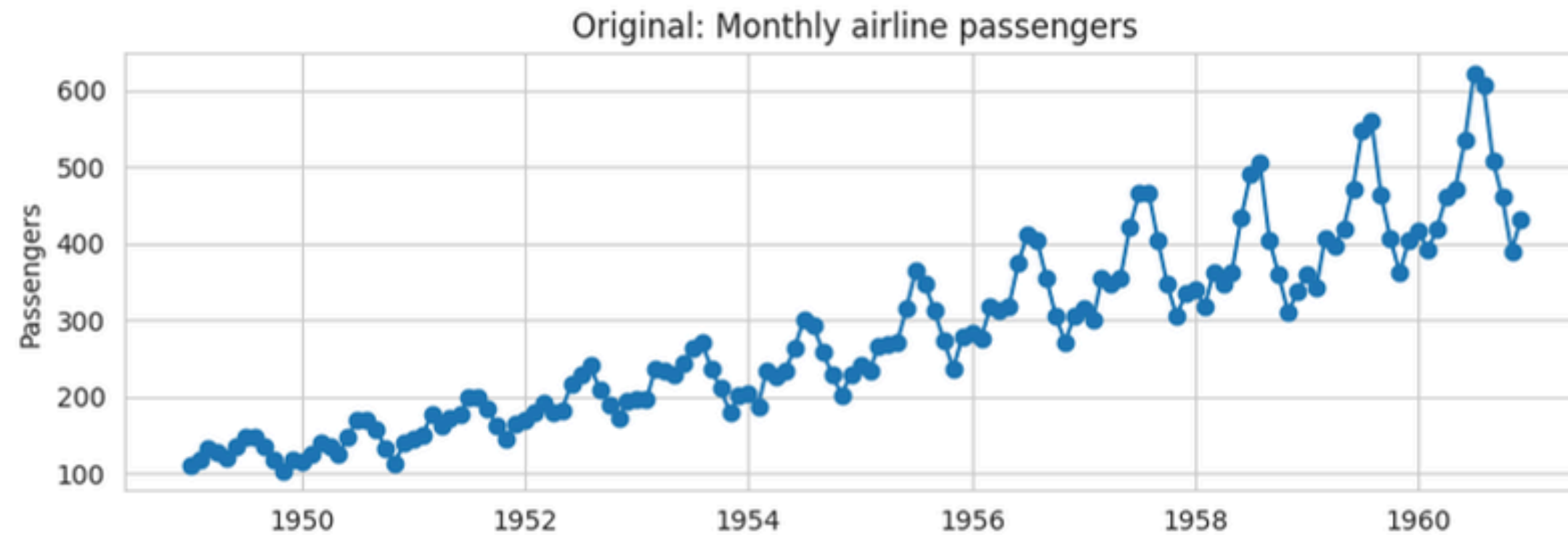
What about this graph between time and temperature?

- As time increases, there is a pattern in the temperature.
- The Temperature is highest in July–Aug and lowest in Feb.
- Jan Temp is very similar to that of Feb.



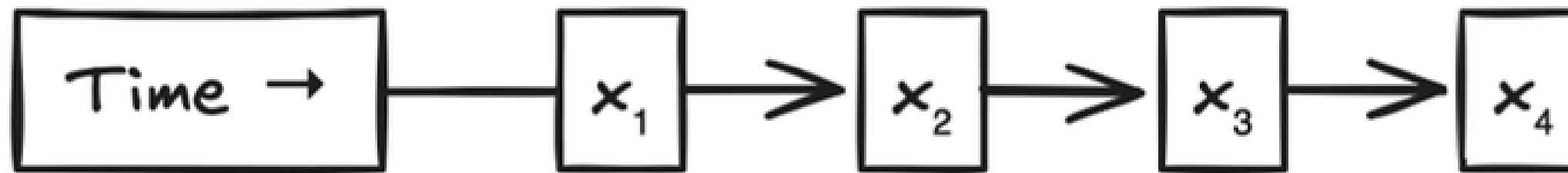
What is Time series?

Time Series Data



- Time series data is a **sequence of data points** collected or recorded at **specific time intervals**.
- Unlike **standard data**, where the **order of observations** may not matter, in time series, the **temporal ordering is critical**.

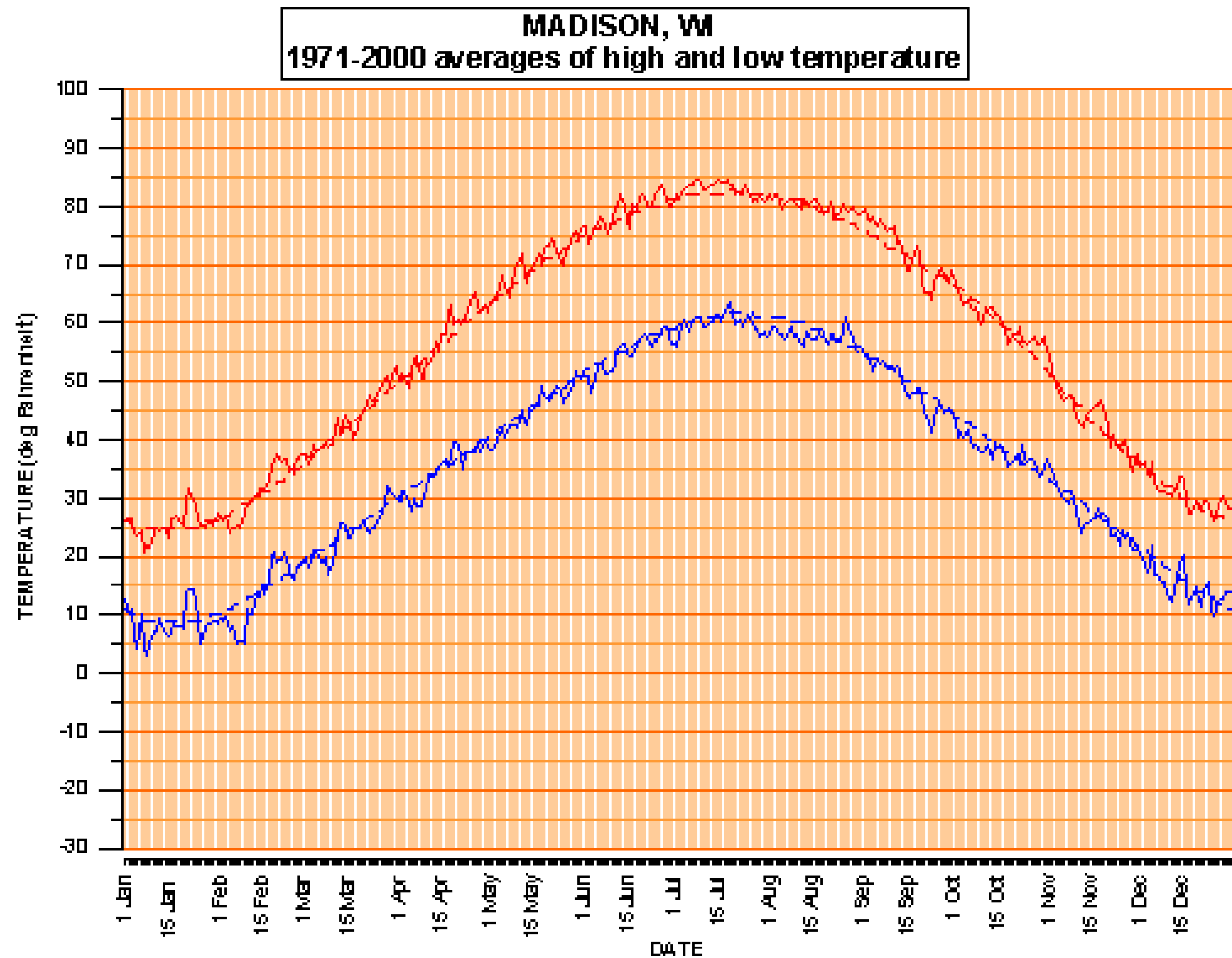
Time Series Data



Values are **indexed** by **time**, and the **order** cannot be **changed**.

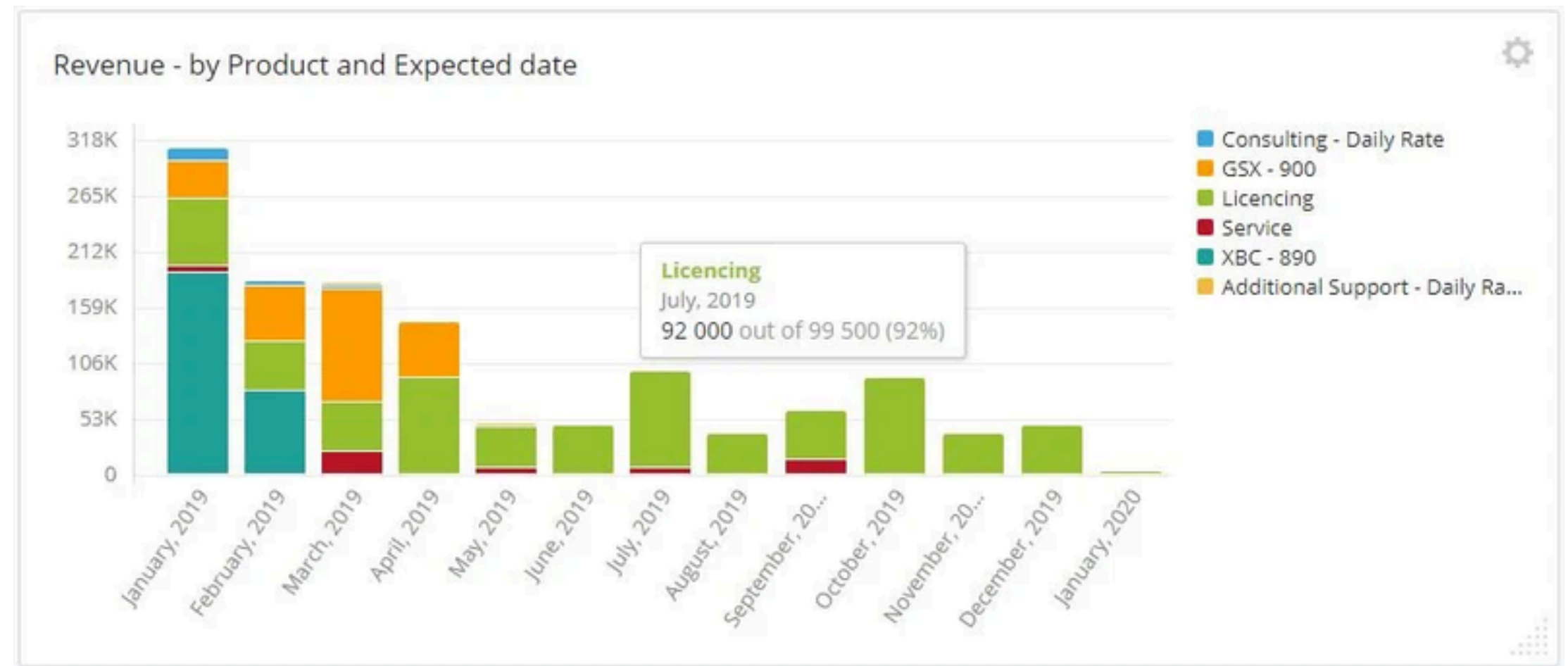
Examples of Timeseries

Weather Data: Daily Temperature



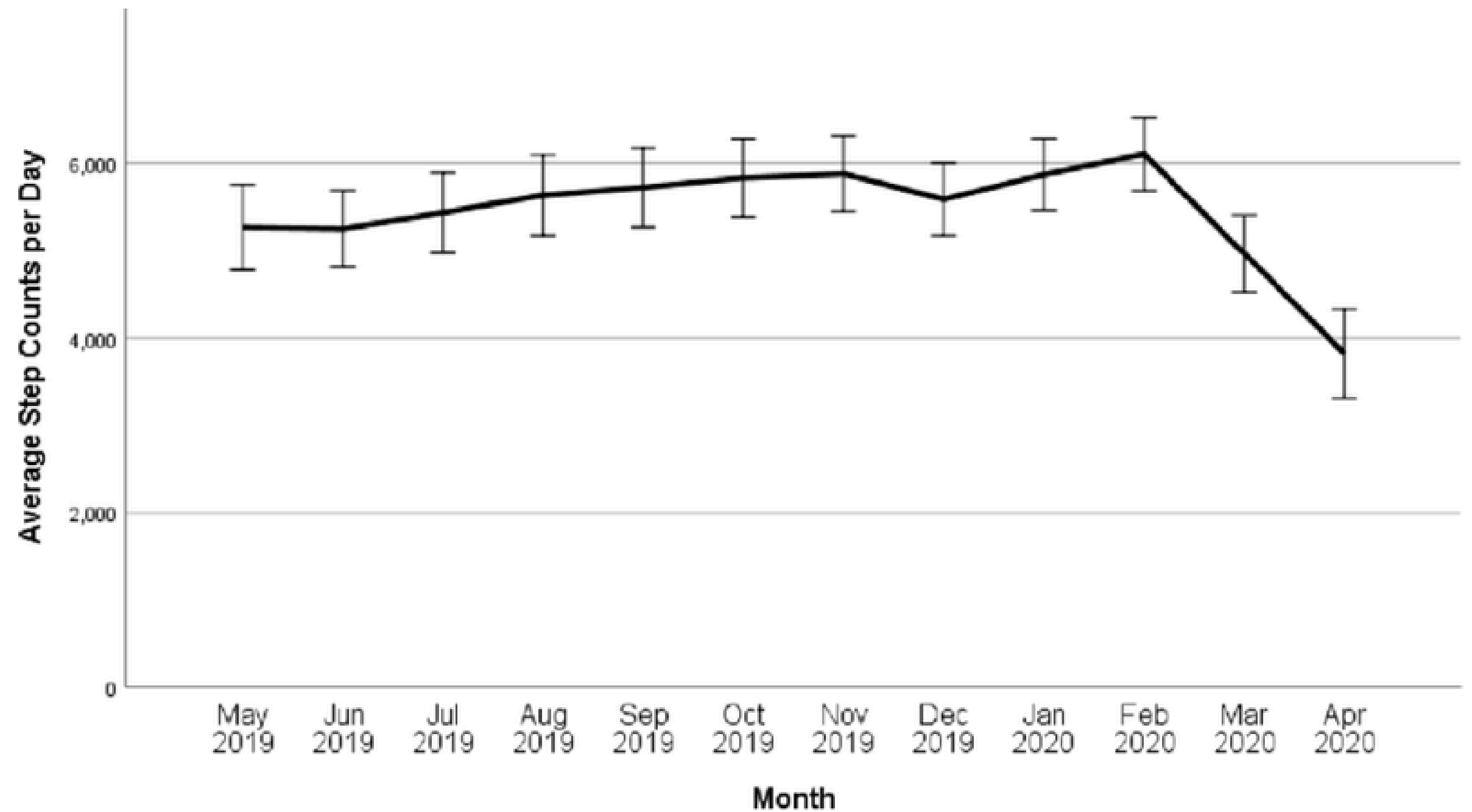
Examples of Timeseries

Sales/Revenue Data: Monthly Sales/Revenue for Products



Examples of Timeseries

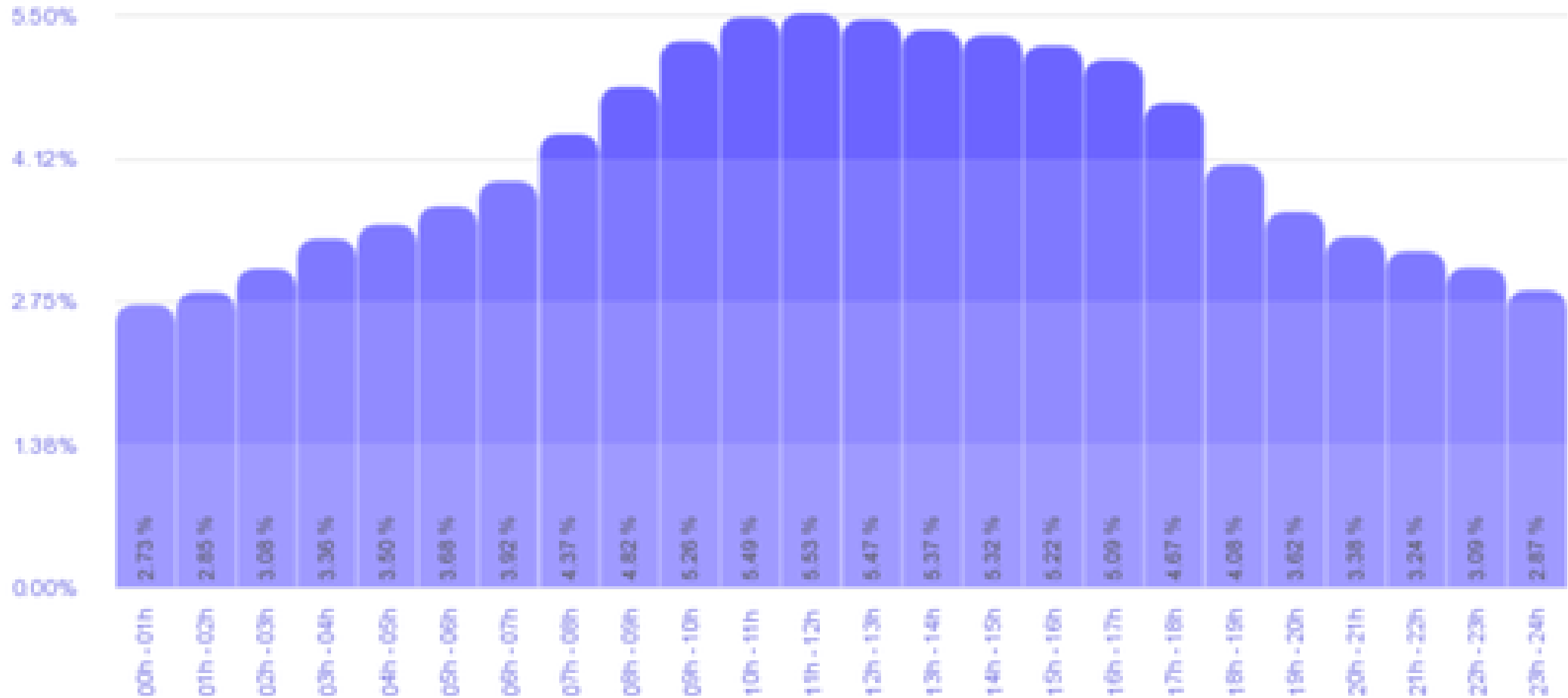
Fitbit Data:
Steps Count per Day



Examples of Timeseries

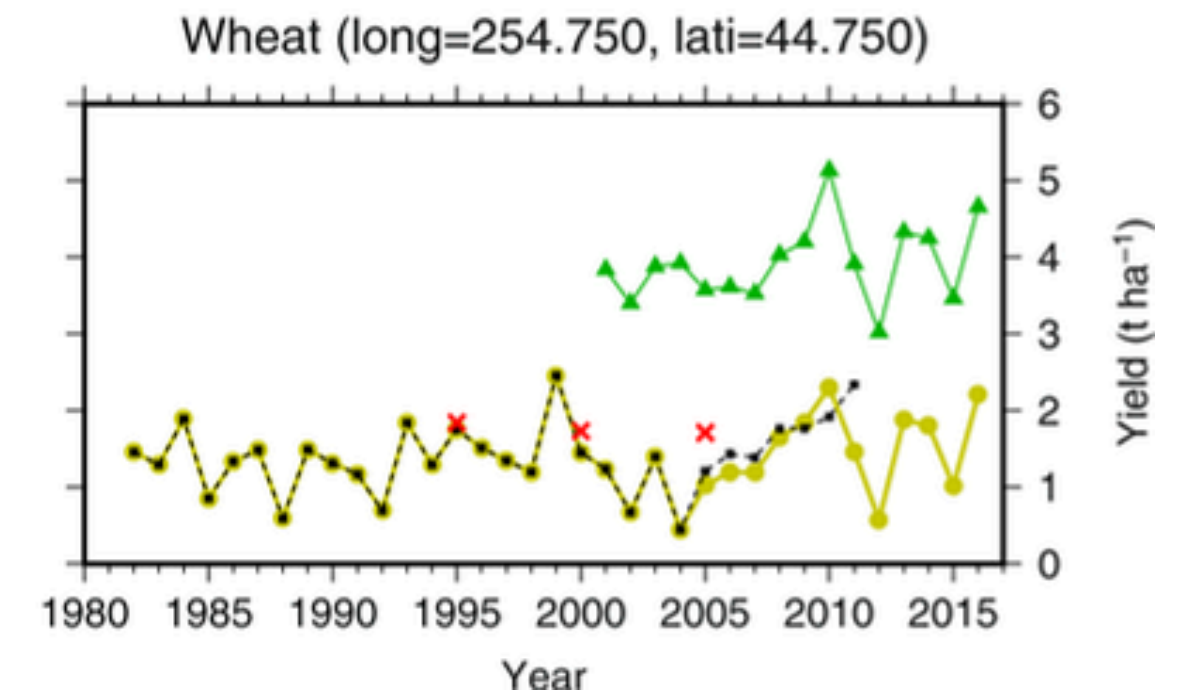
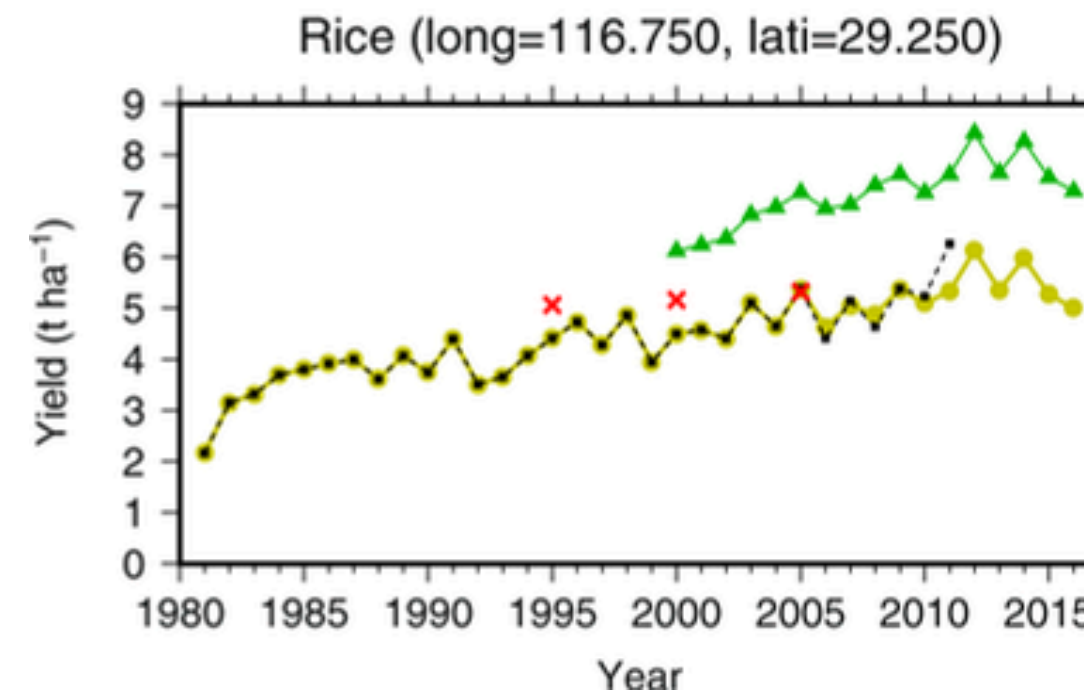
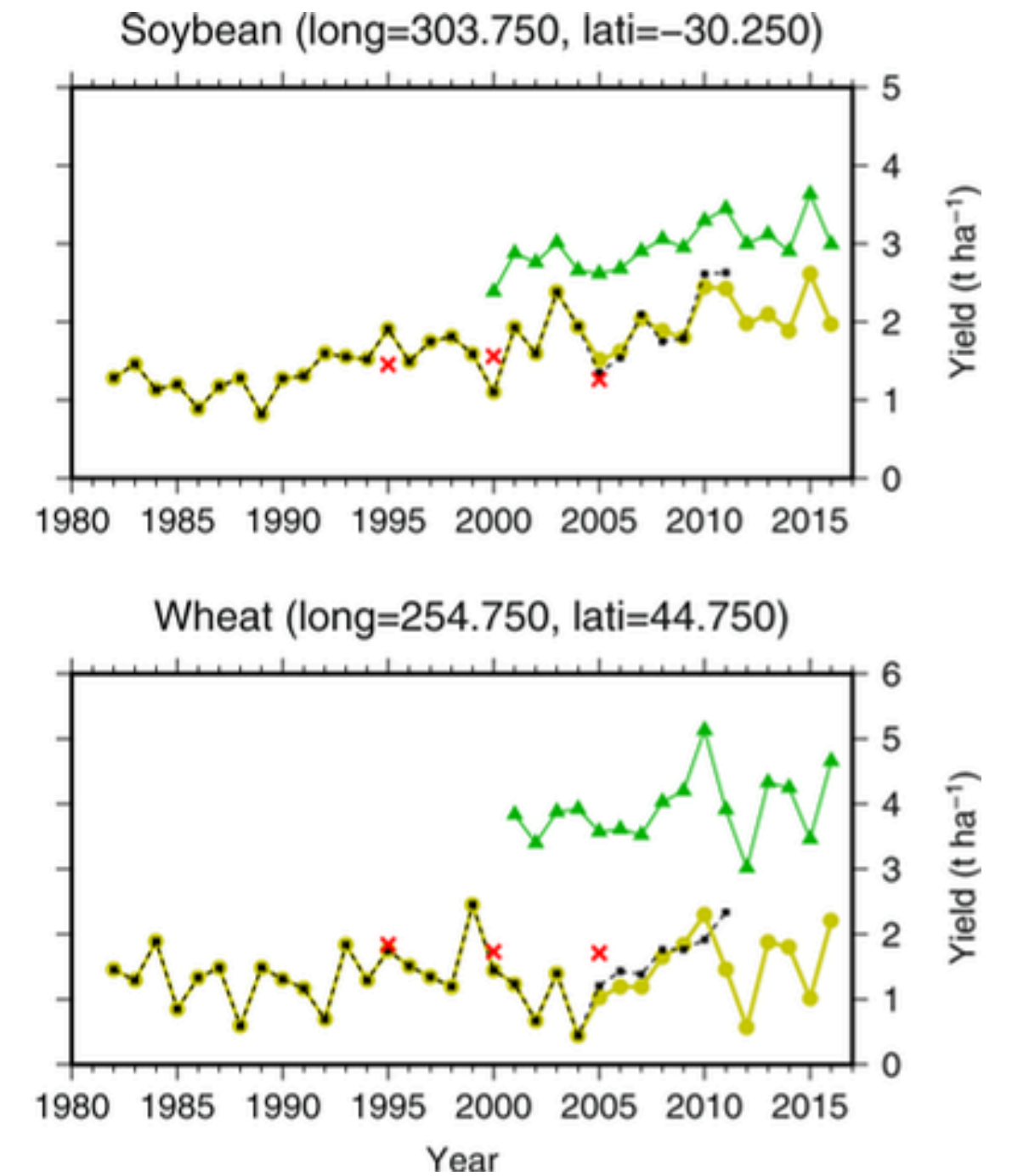
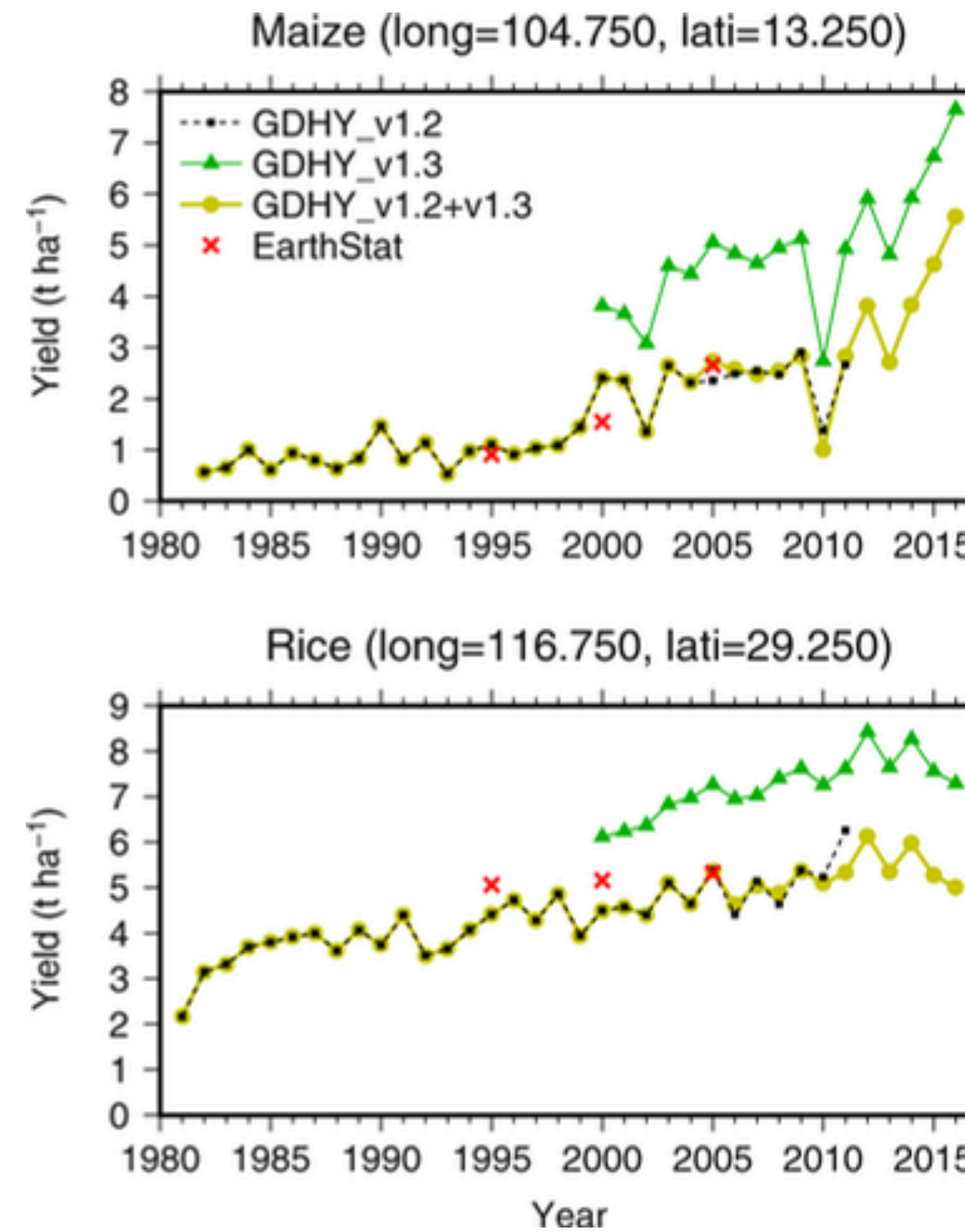
Website Traffic:
Page views per hour/day

Traffic per Hour of the Day
for your local time (GMT -5)



Examples of Timeseries

Crop Yield Over Years

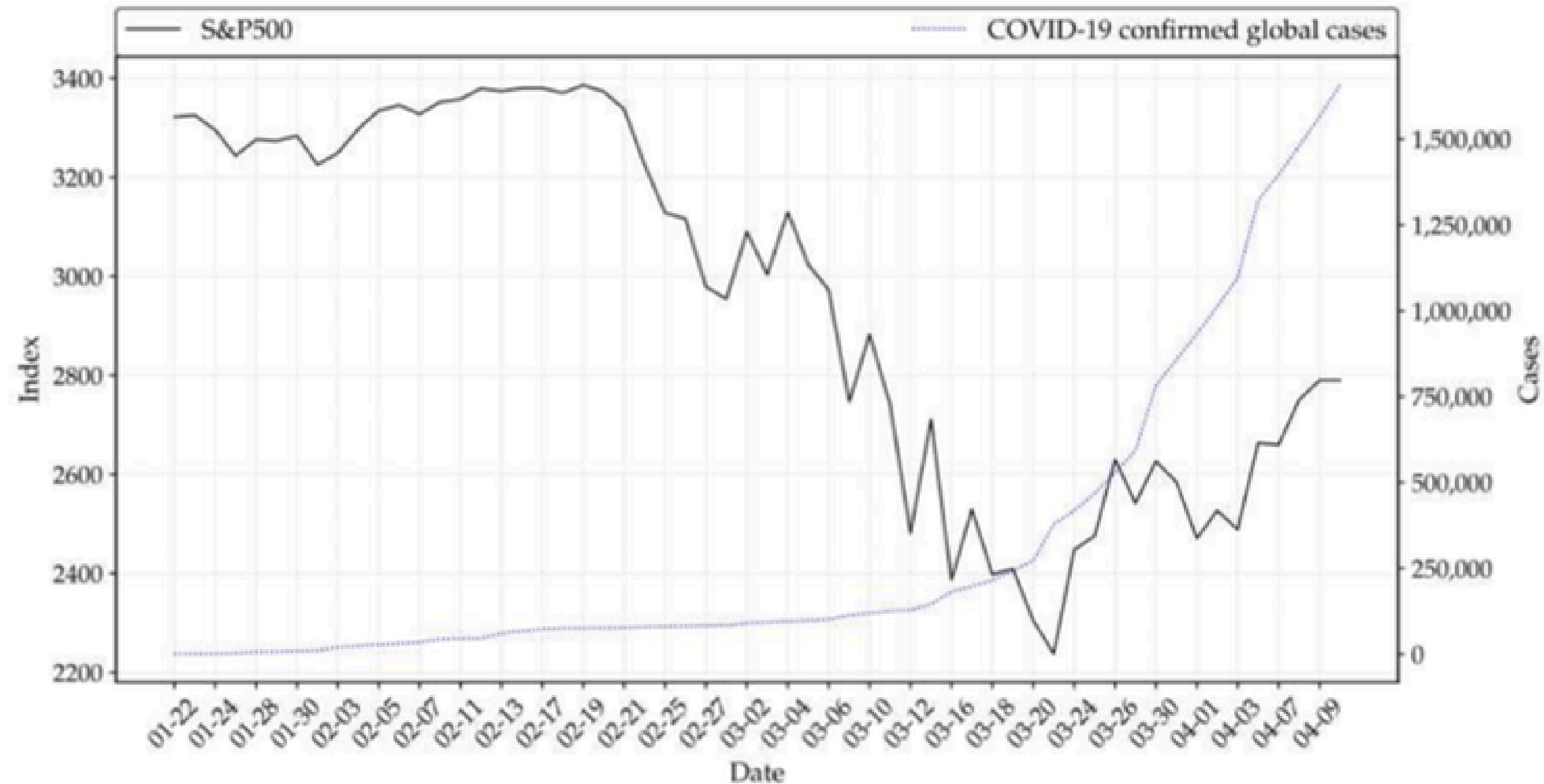


Examples of Timeseries

Stock Price/commodity
Price/Bitcoin etc

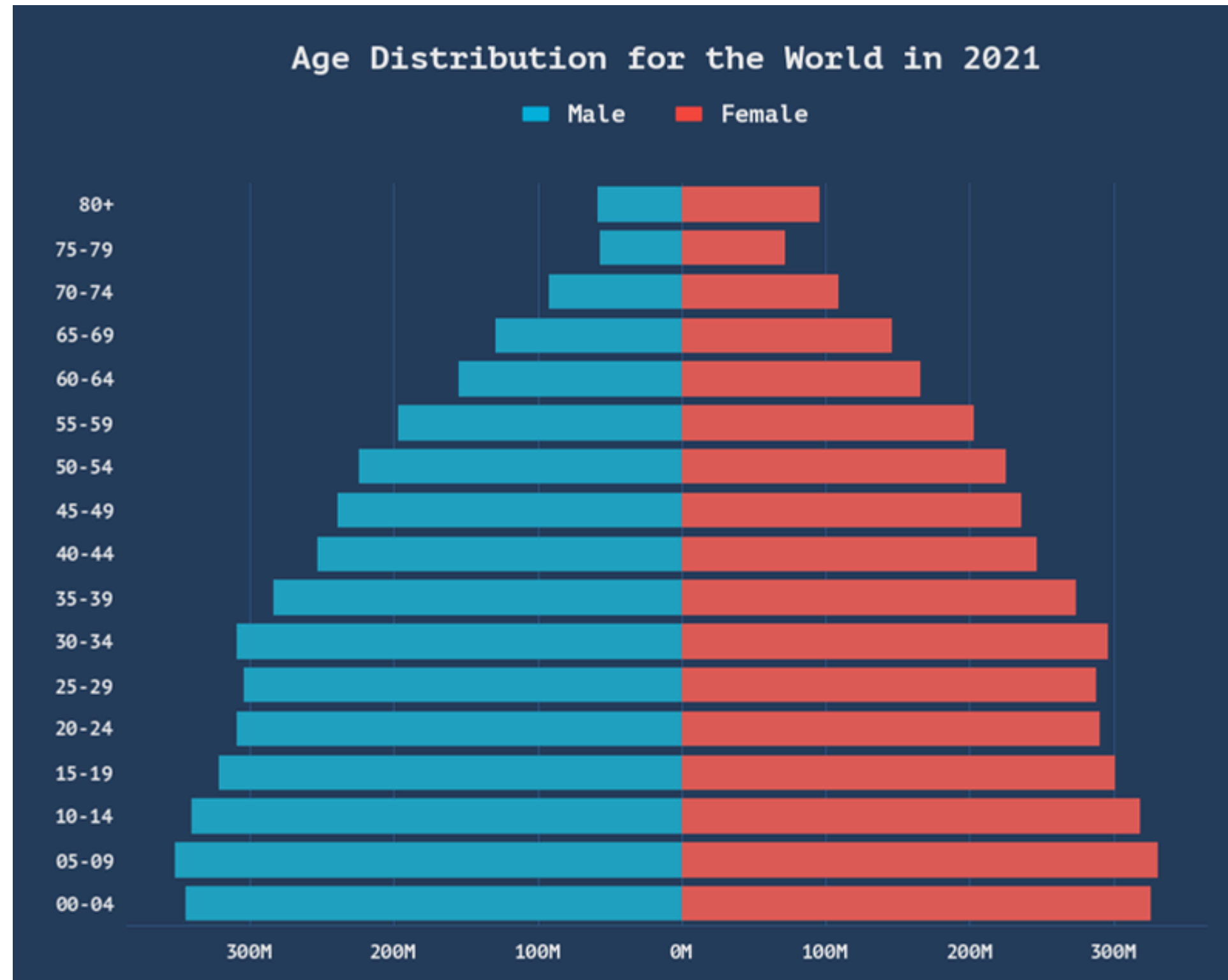


Examples of Timeseries

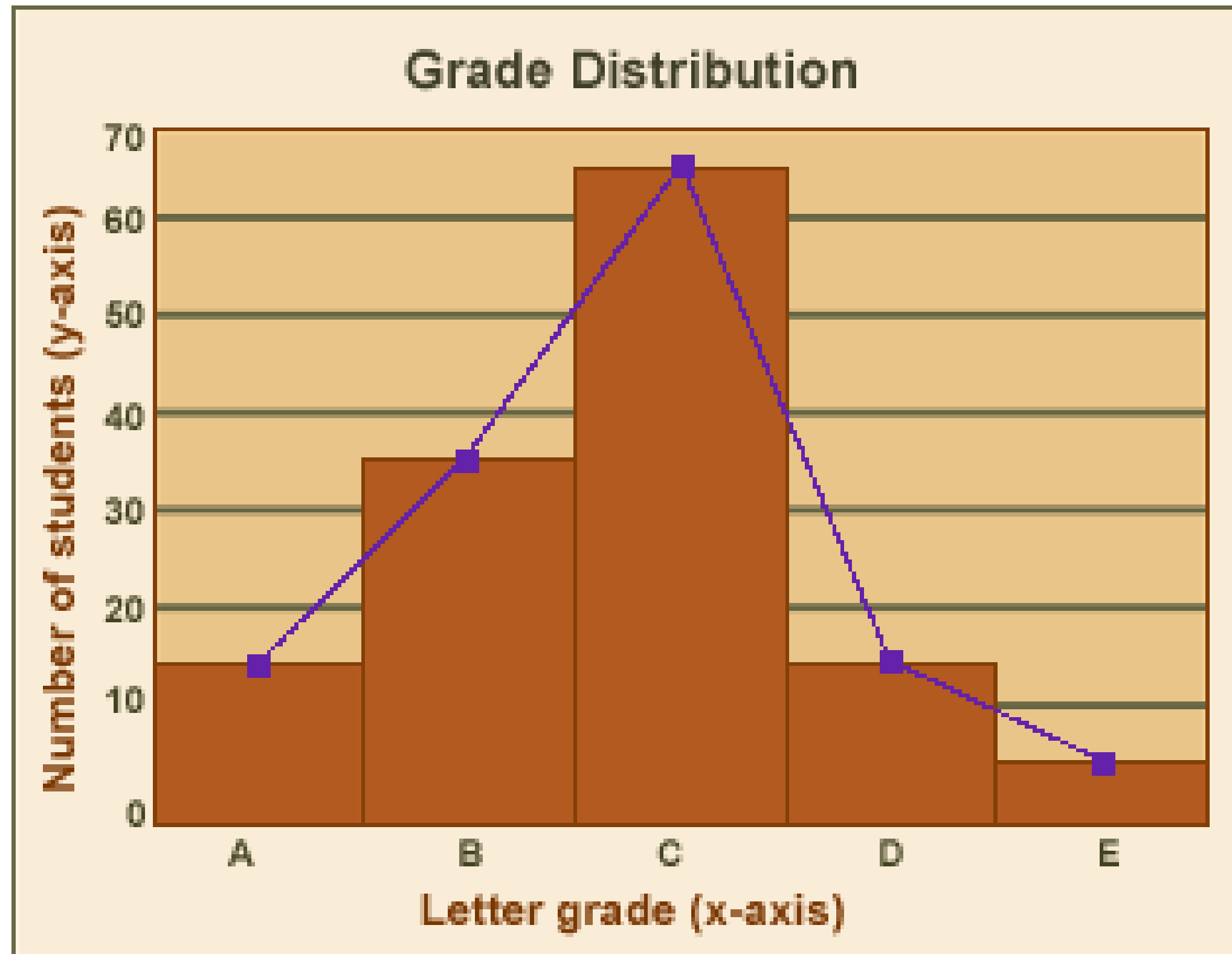


S&P500 vs. COVID-19 confirmed cases (Engelhardt et al., 2020)

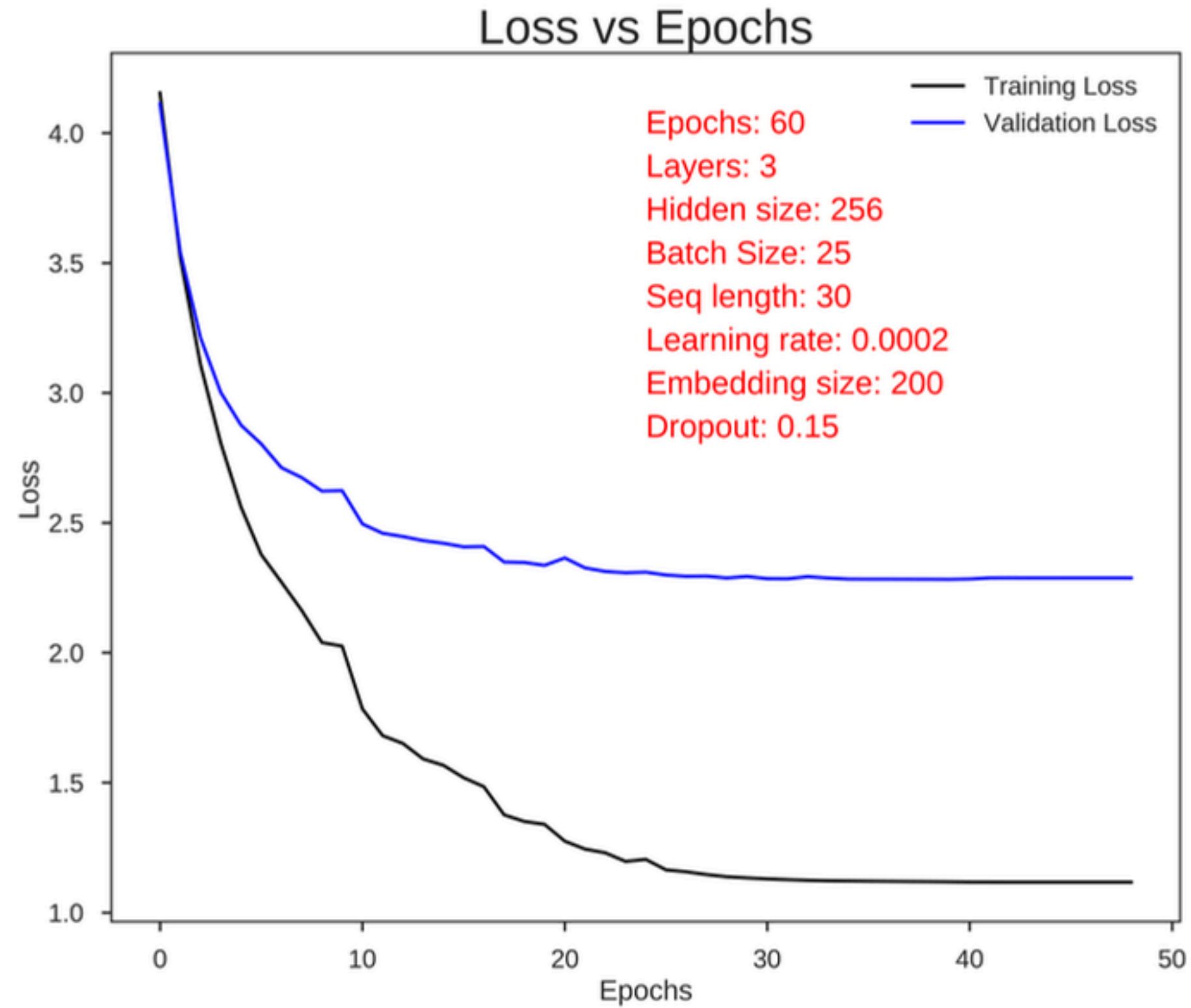
Is this a timeseries?



Is this a timeseries?



Is this a timeseries?



Is this a timeseries?

Student_ID	Submission_Time (minutes)	Exam_Score
S01	35	78
S02	10	92
S03	55	60
S04	20	88
S05	45	70
S06	30	80

Problem

A **factory engineer** records the **number of defects found** after **each production** batch inspection.

Batch Number	Defects Found
1	14
2	11
3	9
4	8
5	6
6	5
7	5
8	4

Problem

A **factory engineer** records the **number of defects found** after **each production** batch inspection.

Is this **dataset** a
time series?

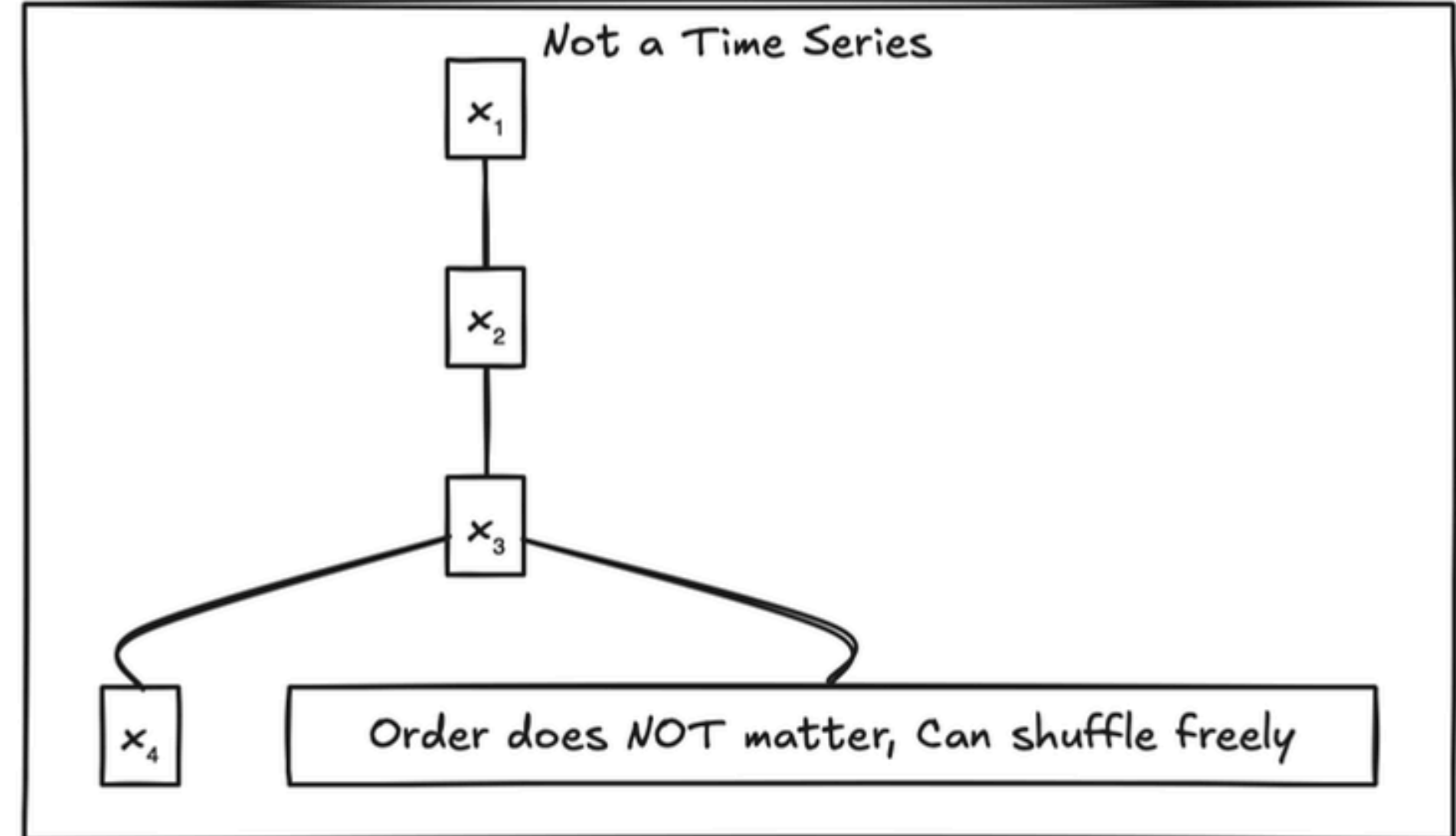
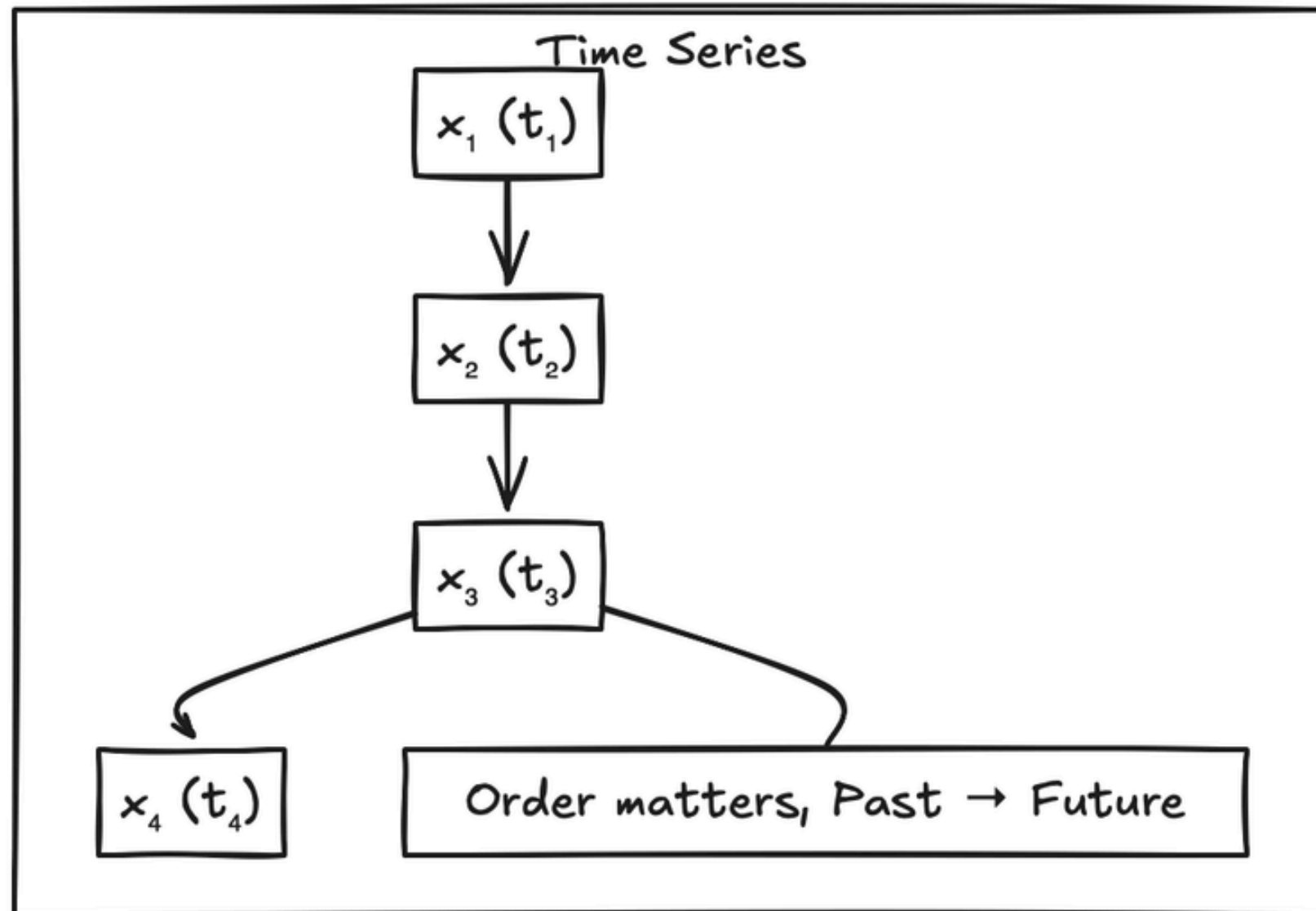
Batch Number	Defects Found
1	14
2	11
3	9
4	8
5	6
6	5
7	5
8	4

Solution

- **Observations are sequential**
- The **order cannot be changed**
- **Past production improvements affect future defect counts.**
- This is a **time series** data.

Batch Number	Defects Found
1	14
2	11
3	9
4	8
5	6
6	5
7	5
8	4

How to identify if data is timeseries?



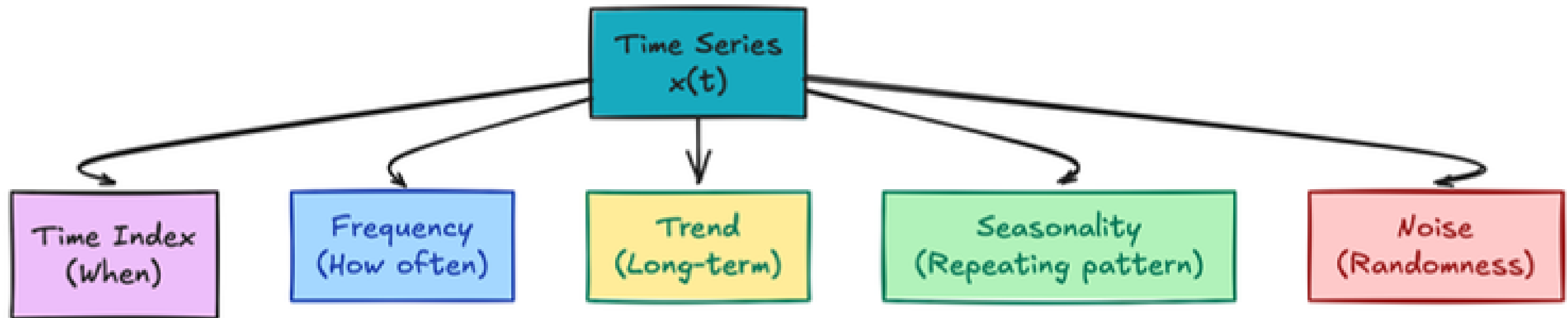
Time series: values are indexed by time, and changing order breaks meaning
Non-time series: values may be ordered, but order carries no information

If you can shuffle the x-axis and nothing breaks, it was never a time series.

All time series are line graphs, but not all line graphs are time series.

Property	Time Series	Looks Like Time Series
X-axis	Time	Anything ordered
Order matters	✓	✗
Temporal dependency	✓	✗
Forecasting possible	✓	✗
Can shuffle x-axis	✗	✓

Component of Time Series

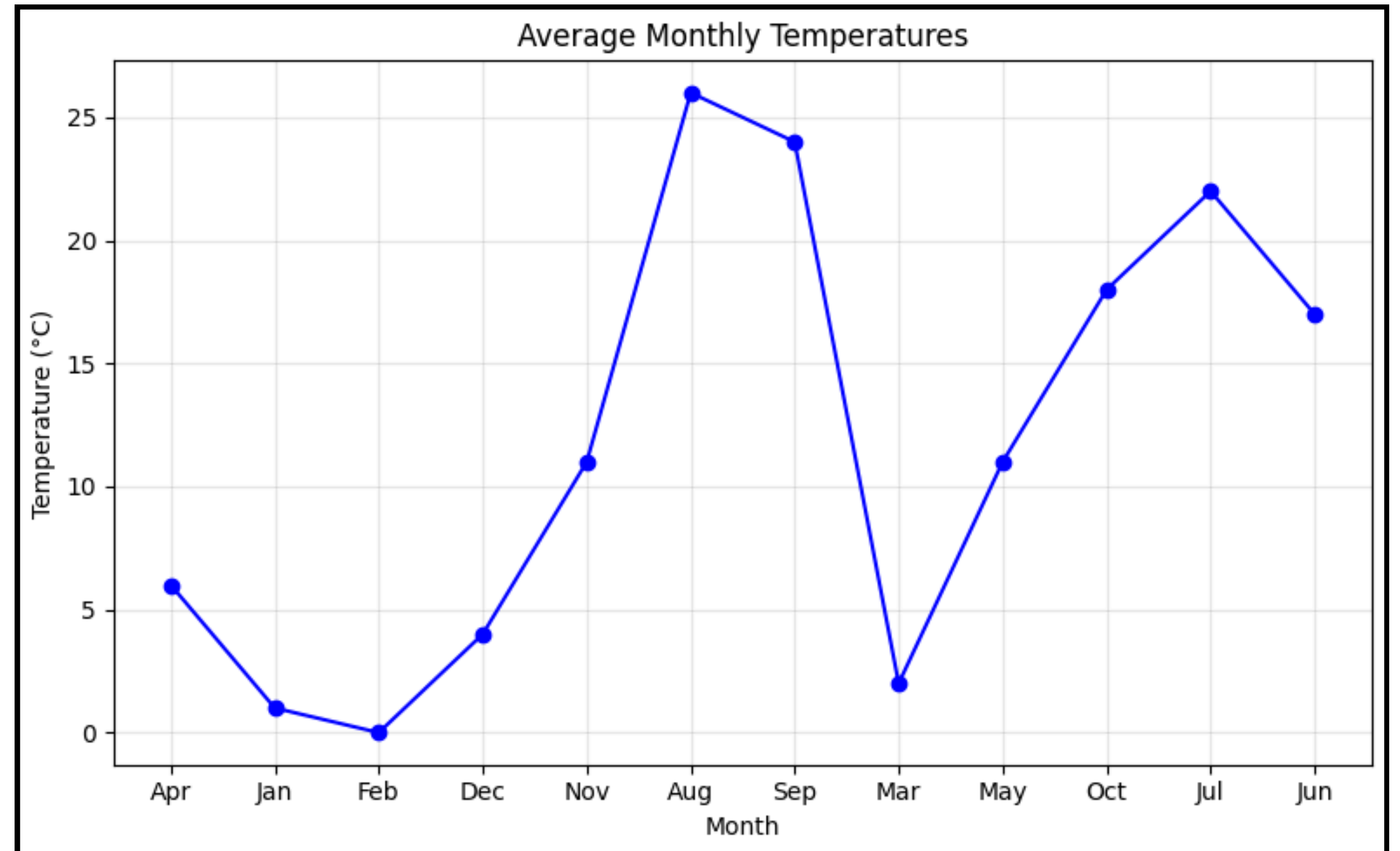


$$\mathbf{x(t)=Trend + Seasonality + Noise}$$

Time Index and Frequency (structure),
Trend, Seasonality, Noise (behavior)

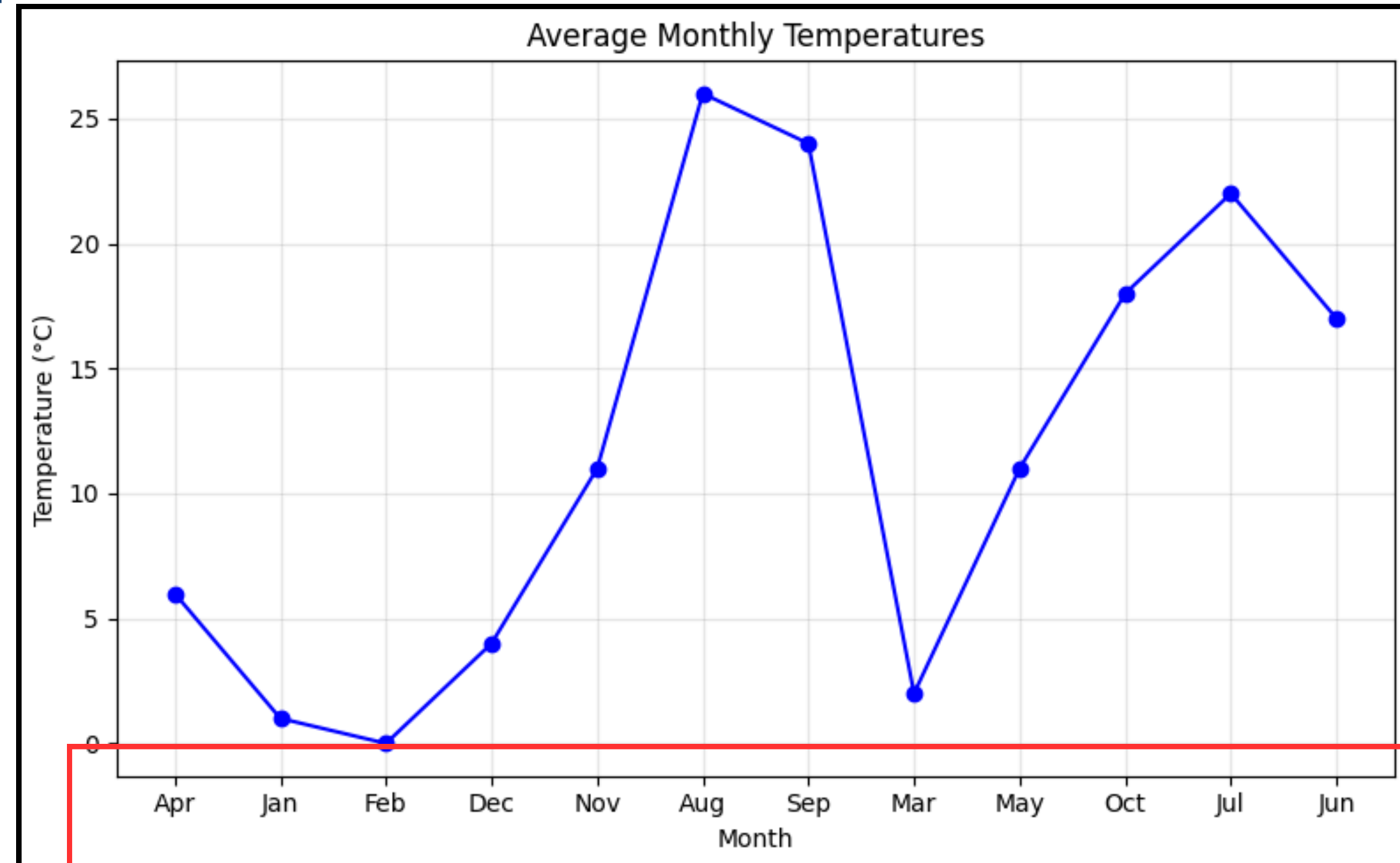
What will happen if we arrange the data at random?

- What will happen if we arrange the data at random?



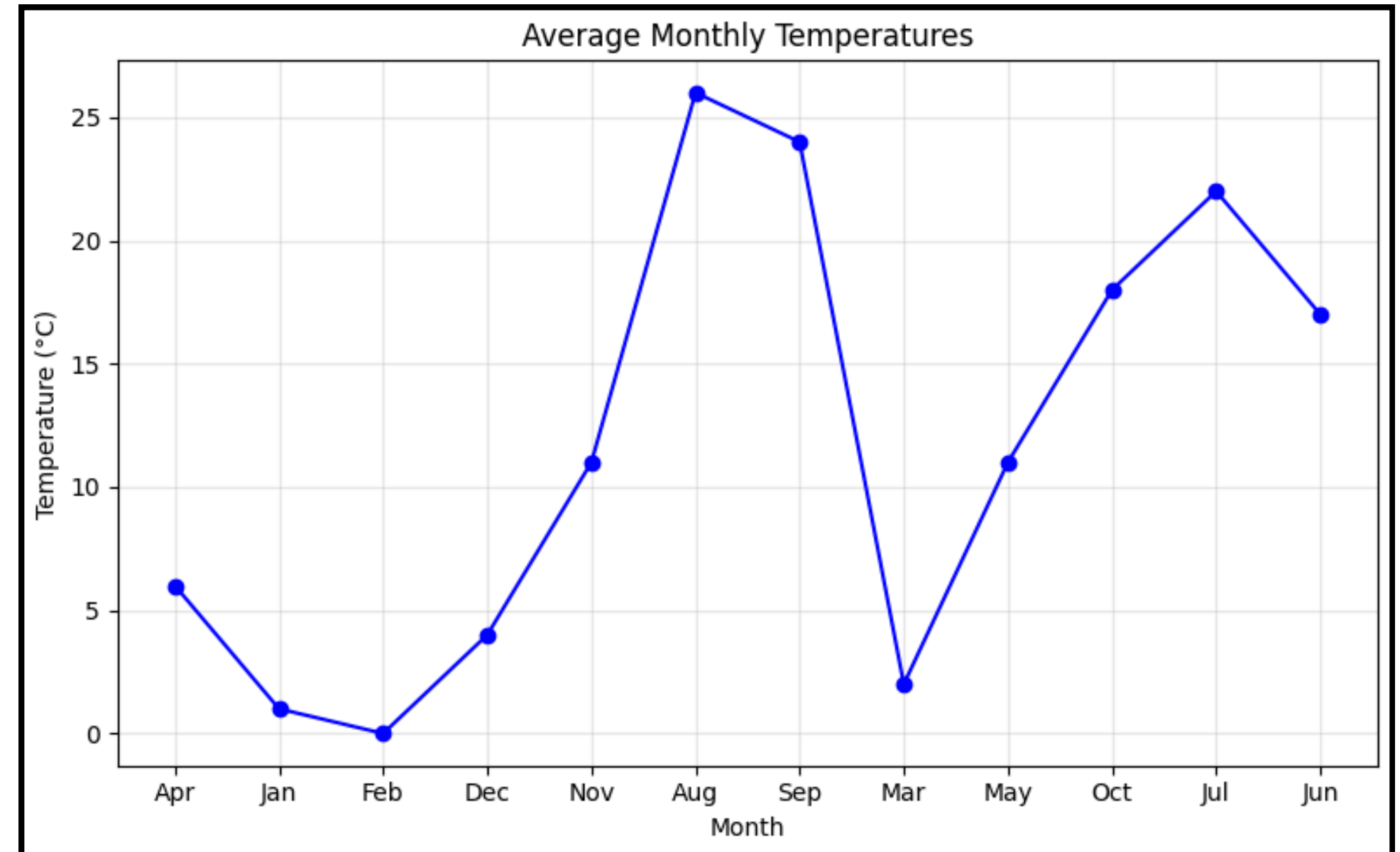
Time Index

- The time index tells us WHEN each point was recorded.
- Without correct time stamps:
 - order becomes unclear
 - patterns break
 - interpretation becomes unreliable
- This is why shuffling the data earlier destroyed meaning — the time index was violated



Time Index

- Time Index allows us to:
 - detect gaps
 - compare events
 - measure spacing between observations



If we record something more often or less often, how does the pattern change?

Frequency

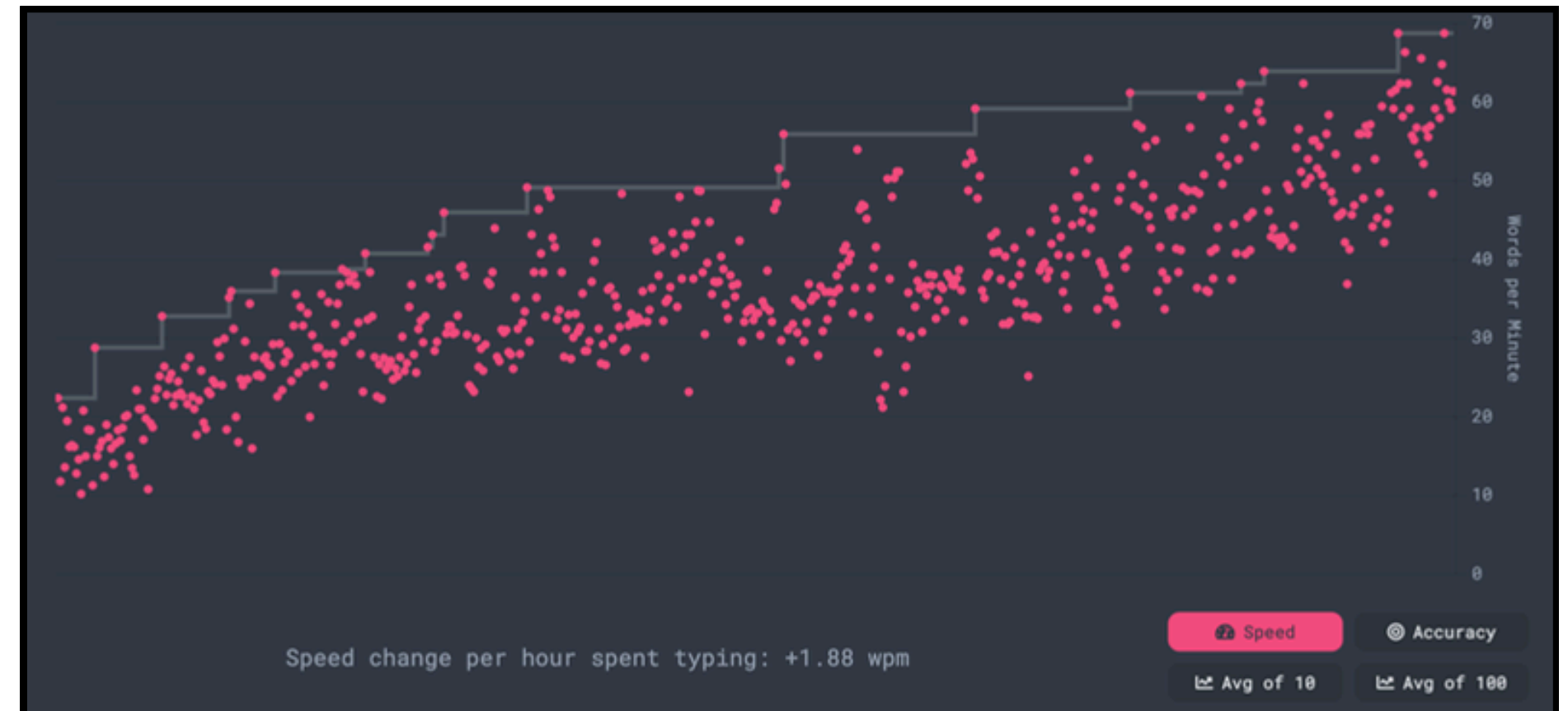
- Frequency = time gap between observations
- **Examples:**
 - **Yearly (Inflation):** Focuses on long-term trends; ignores temporary noise.
 - **Monthly (Temperature):** Reveals seasonal cycles; smooths out daily variance.
 - **Daily (COVID):** Tracks short-term fluctuations and weekly patterns.
 - **Hourly (Traffic):** Shows intraday peaks and valleys (e.g., rush hour).
 - **Per-second (IoT):** Captures instantaneous, critical state changes.

Frequency

- How frequency affects meaning:
 - Higher frequency → more detail, more noise
 - Lower frequency → smoother, but hides patterns

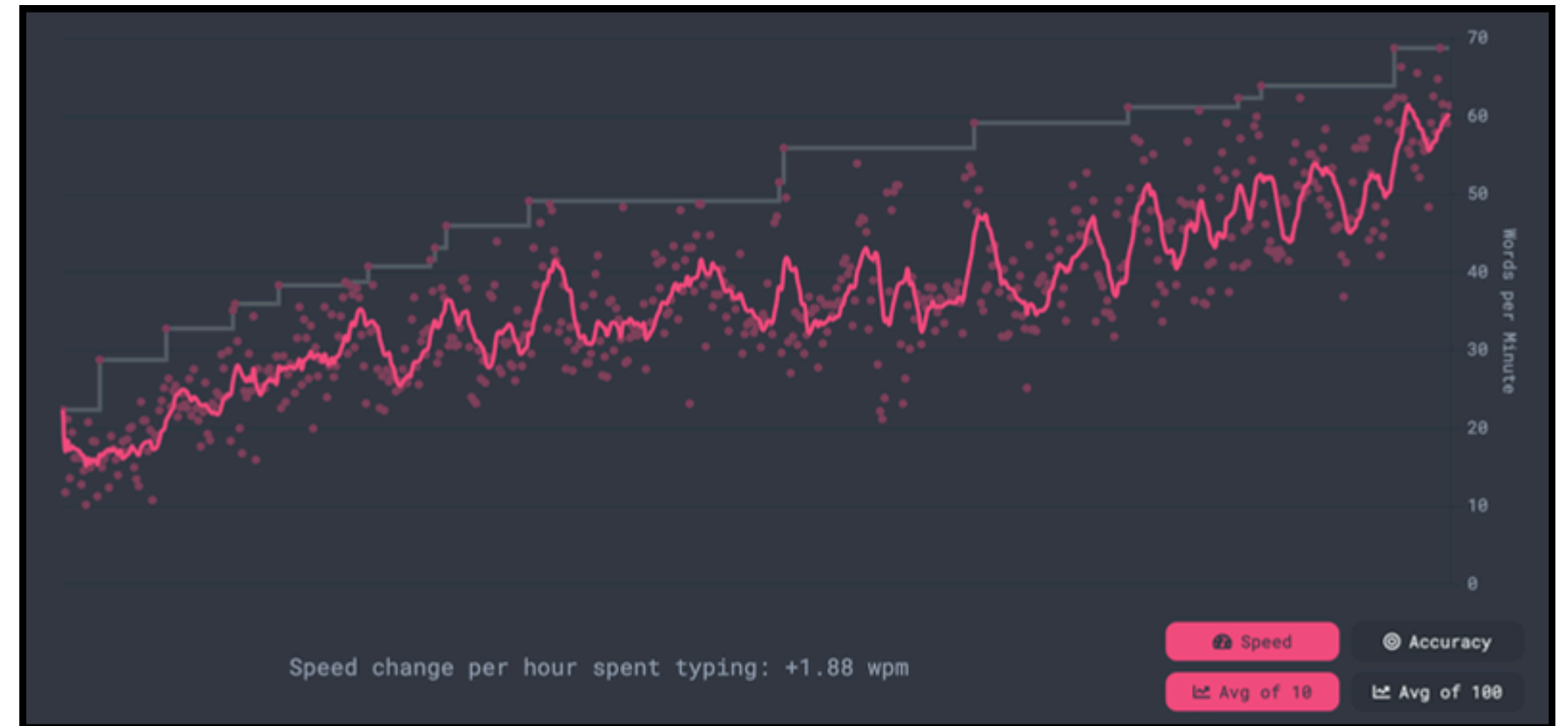
Frequency

- This graph from Typing Monkey Website shows Shobhit Sir's typing progress.
- The **x-axis** represents **test timestamps**, and the **y-axis** shows **typing speed** in words per minute.
- Below are two buttons: one for the average of the **last 10 tests**, and the other for the average of the **last 100 tests**.



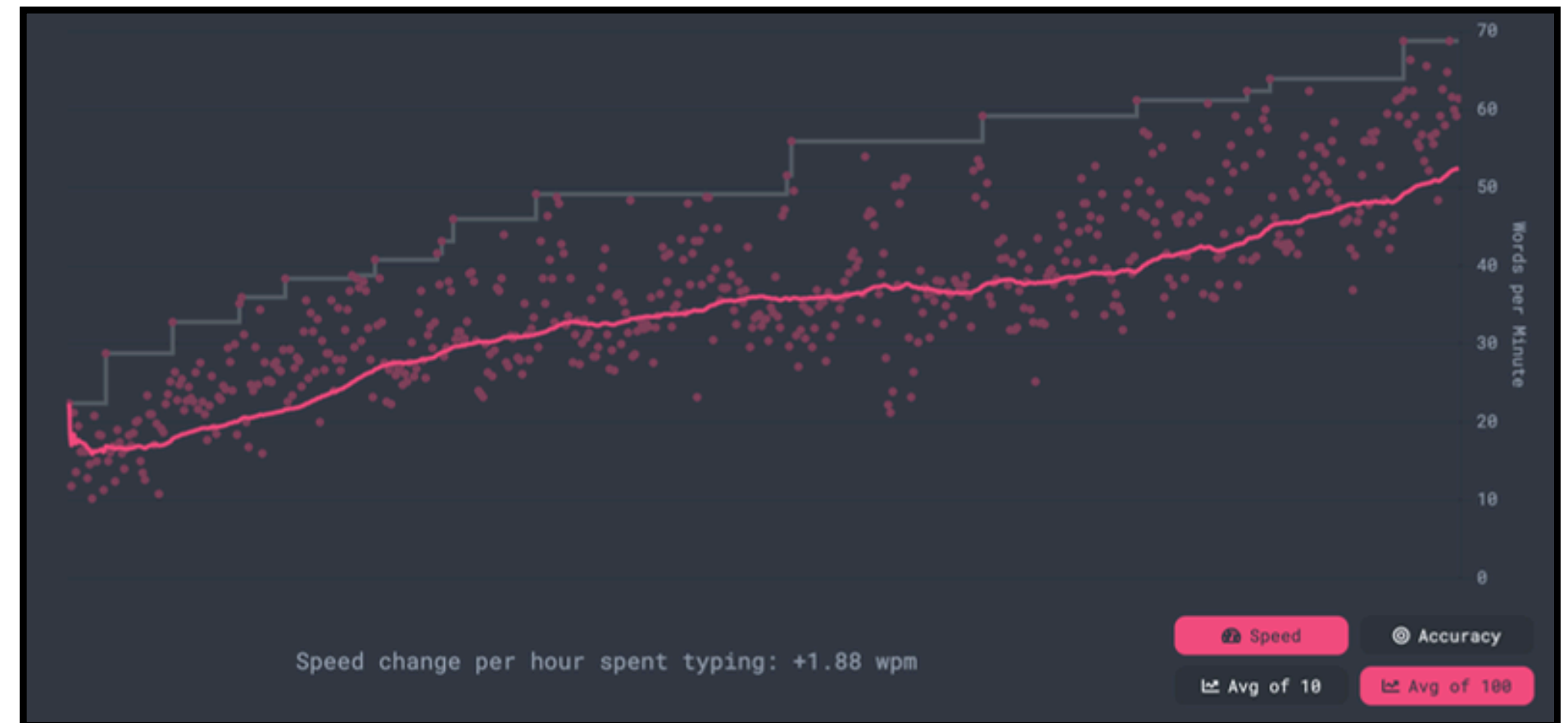
Frequency

When we click the 'Avg of 10' button, it displays **high-frequency** time series data.



Frequency

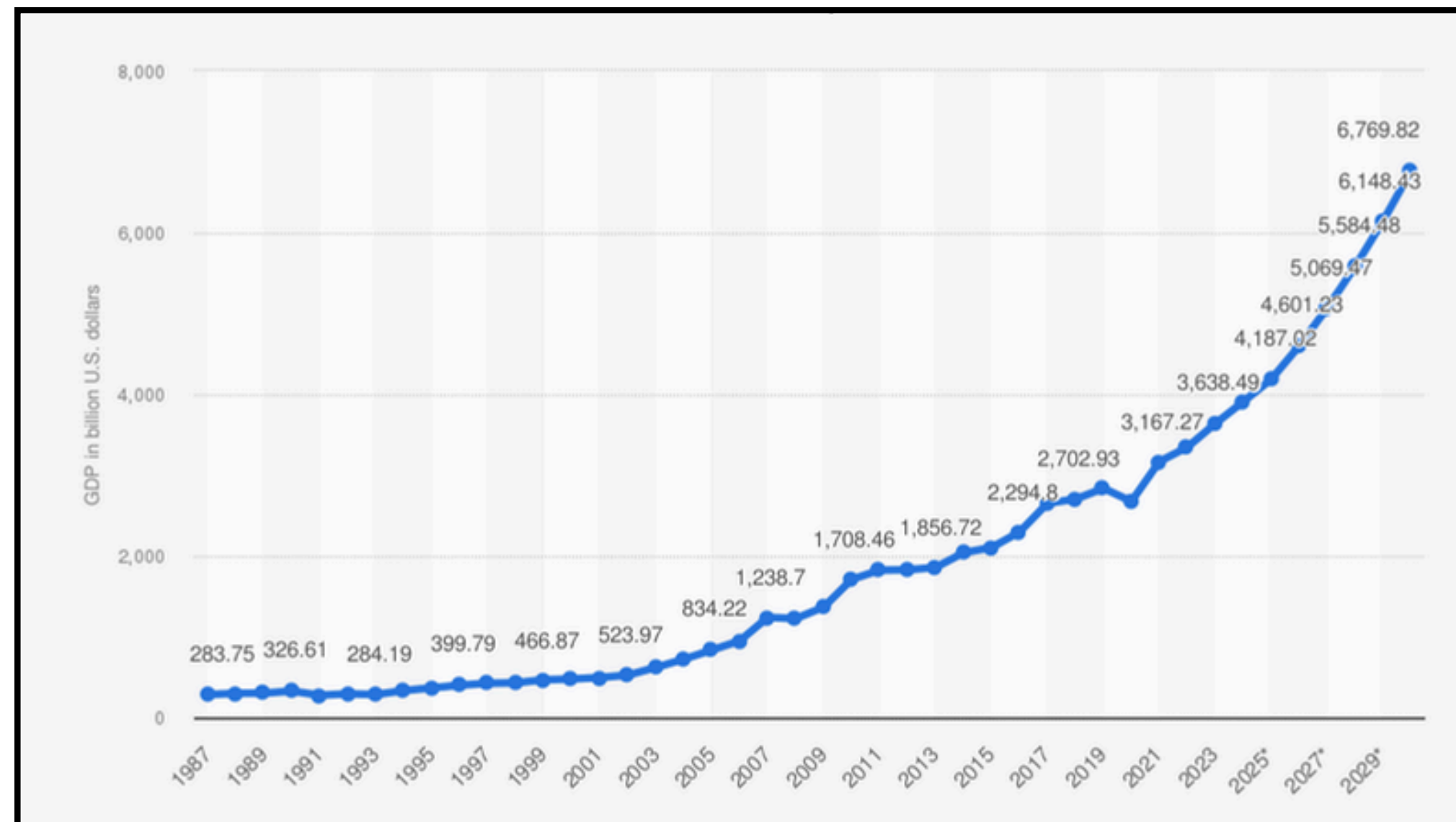
When we click the 'Avg of 100'
button, it displays **low-frequency**
time series data.



Frequency

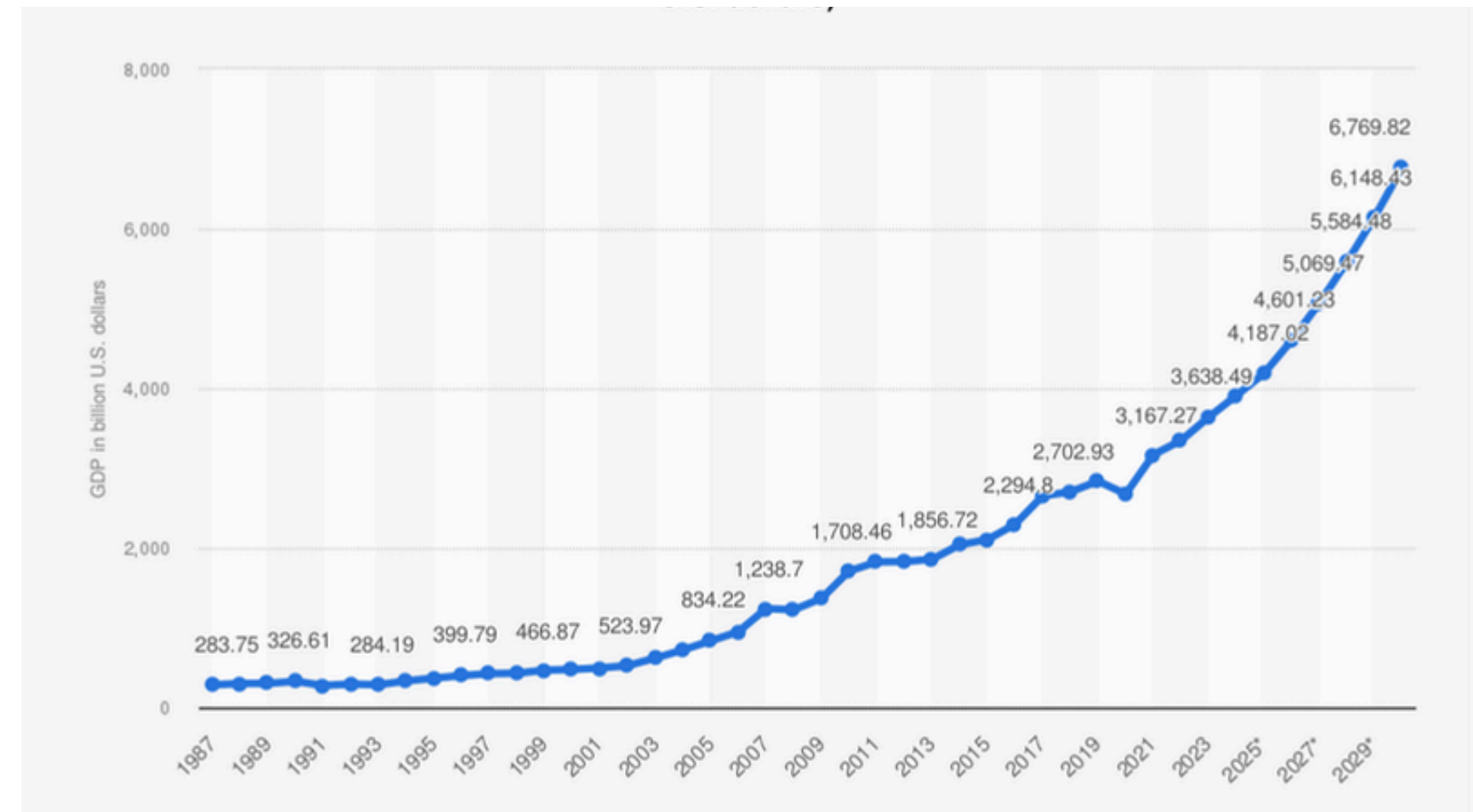
Frequency	Best For...	Shows	Hides
Yearly	Policy & History	Decades of trends	Seasons & Specific events
Monthly	Planning & Climate	Seasons (Summer vs Winter)	Daily weather spikes
Daily	News & Reaction	Weekly habits (Weekends)	Hour-by-hour shifts
Hourly	Operations & Logistics	Rush hours & Peaks	Instantaneous glitches
Per-Second	Safety & Machines	Immediate failures	Long-term drifts

Ignore the small ups and downs... what overall direction do you see?



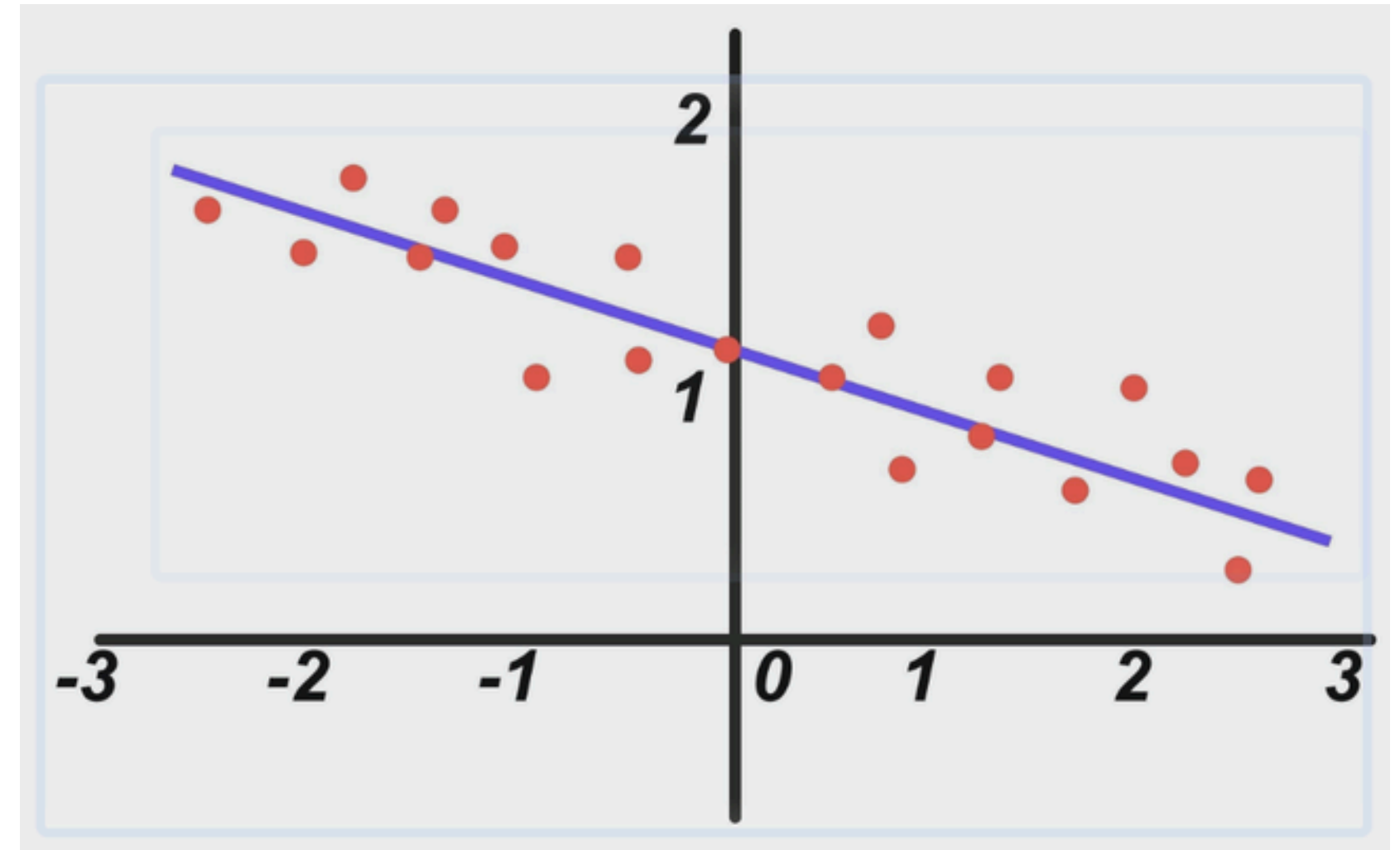
Trend — The long-term movement

- **Trend** = the general direction over long time periods
- **Could be:**
 - Increasing
 - Decreasing
 - Flattening
 - **The “big picture” hidden under local wiggles**

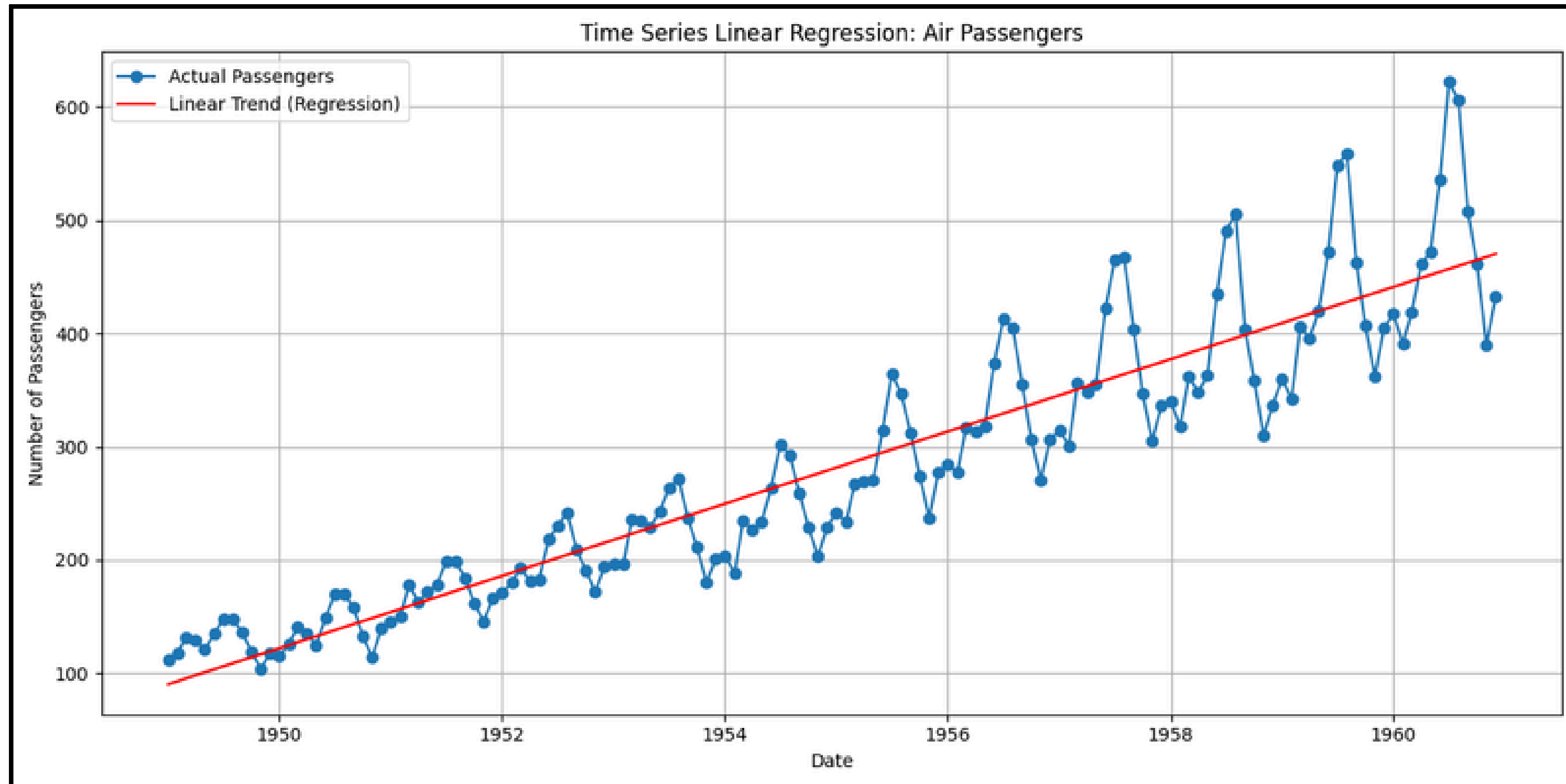


Linear Trends

- **Constant Rate:** The value increases or decreases by a fixed, unchanging amount for every unit of time (e.g., saving exactly \$100 every month).
- **Straight Line:** When plotted on a graph, the data follows a straight diagonal path; it never curves, accelerates, or bends.
- **Predictability:** Because the speed of change is stable, it is the simplest pattern to forecast —you just extend the straight line.

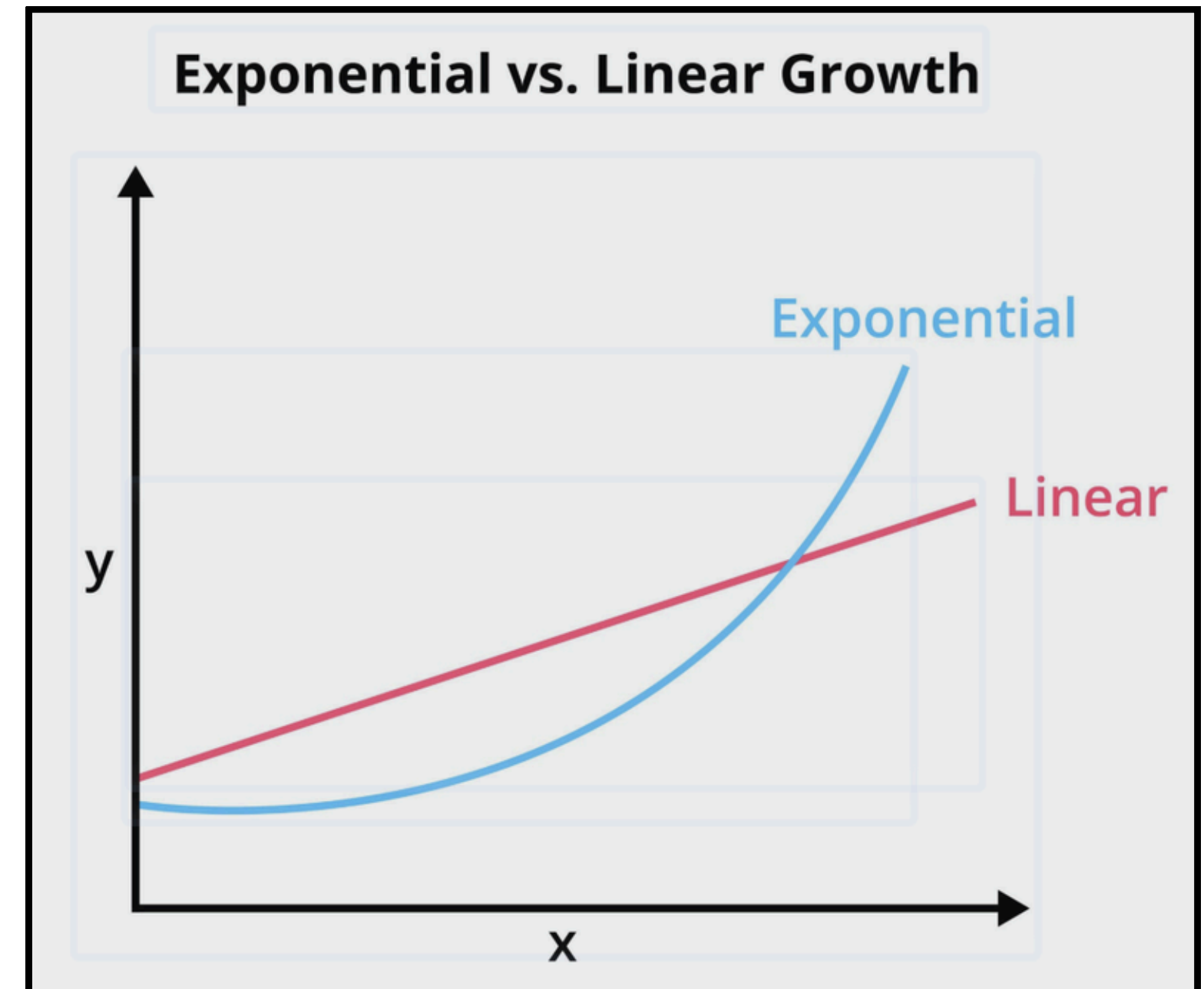


Real life example of Linear Trend

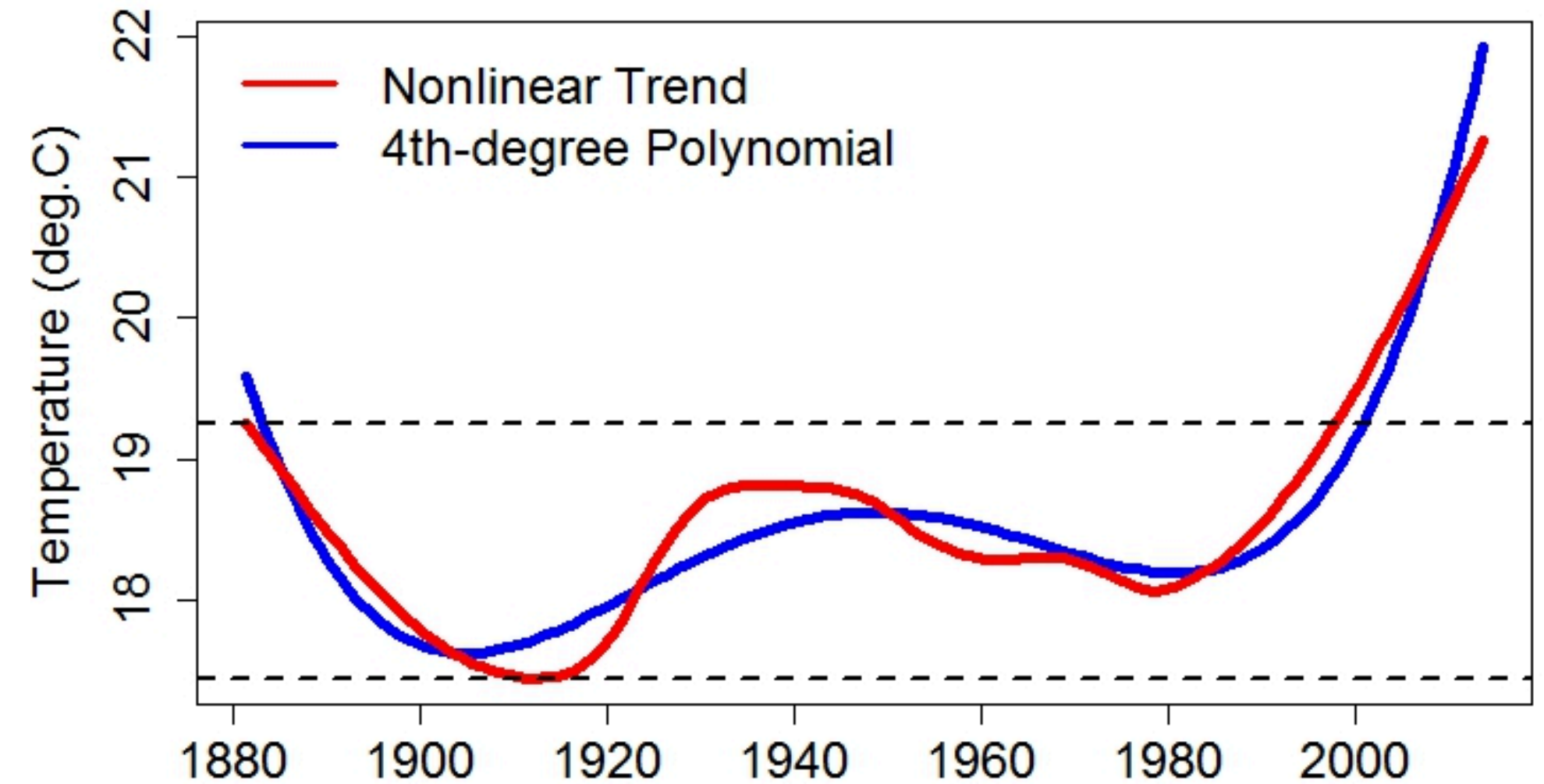
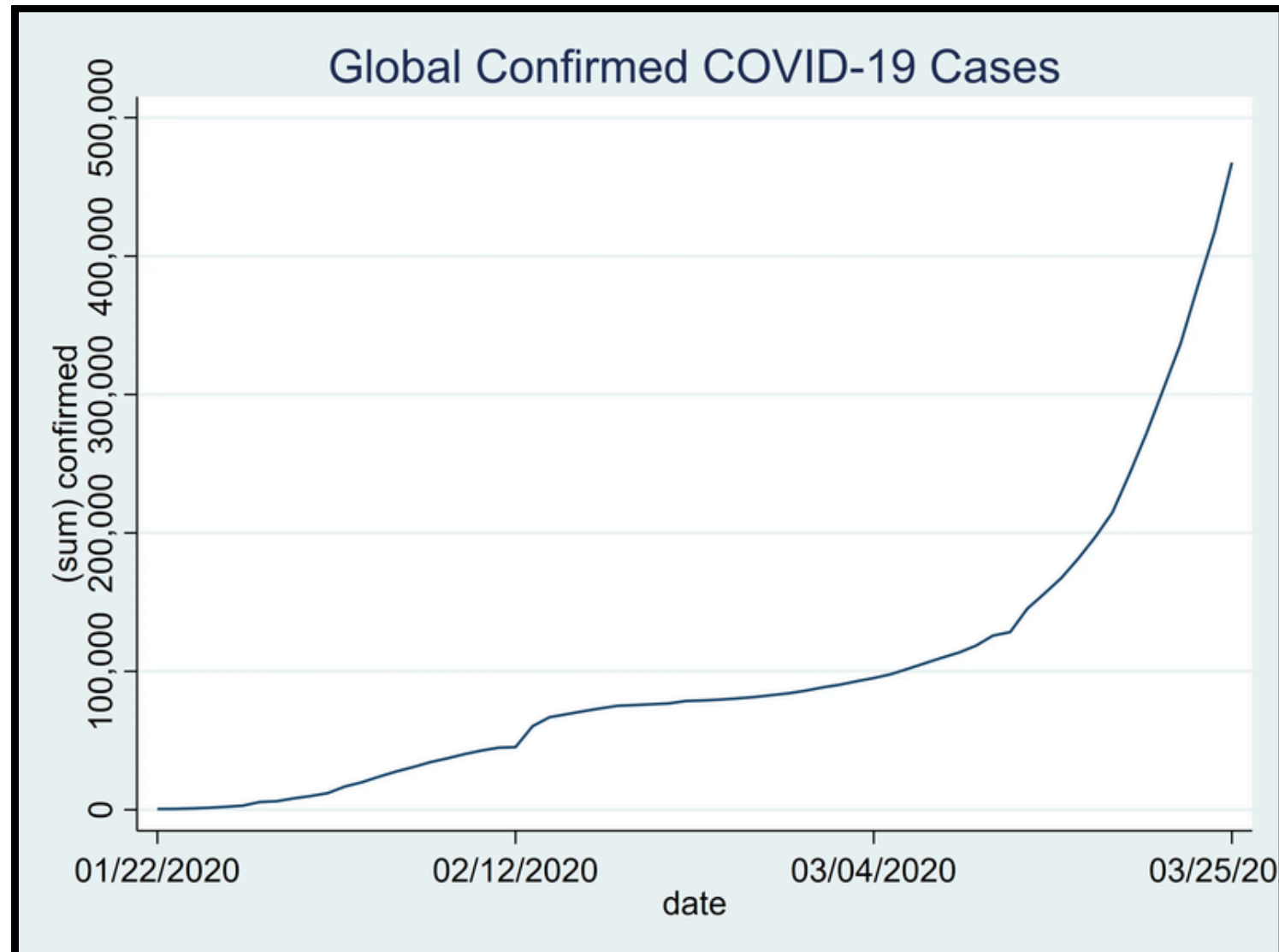


Non-Linear Trends

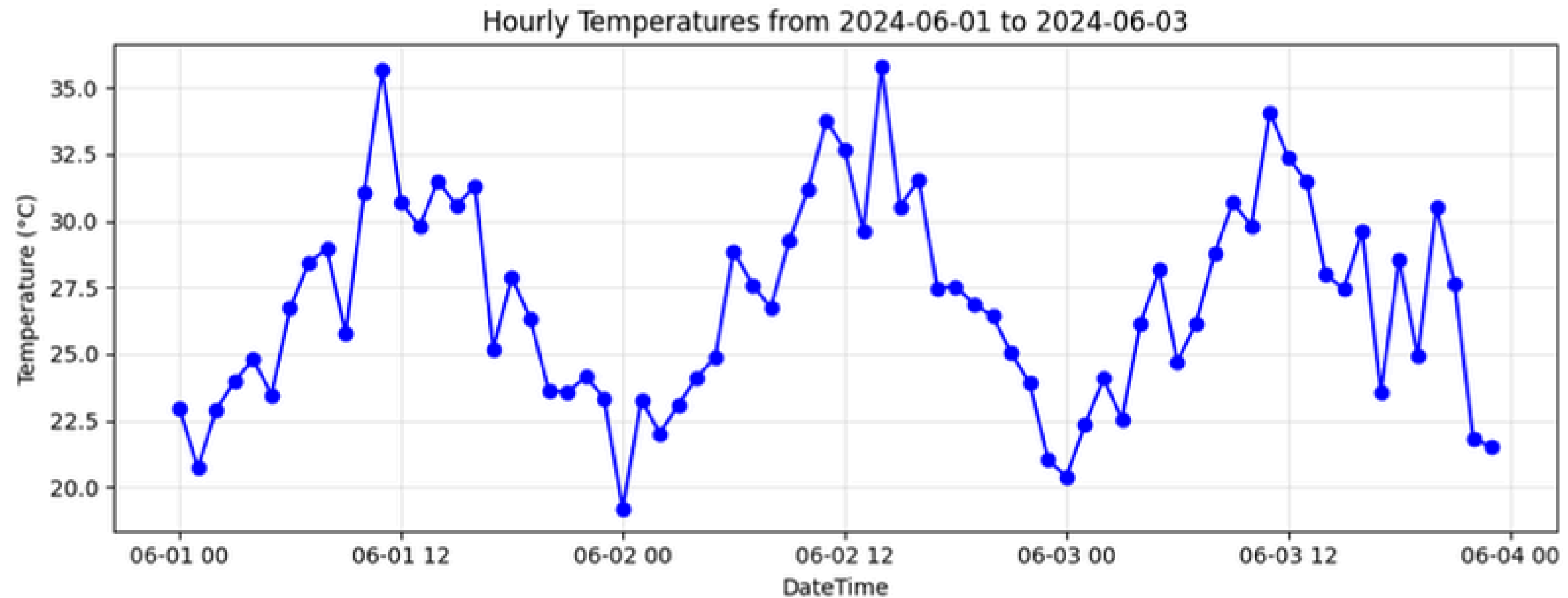
- **Variable Rate:** The speed of change is not constant; it accelerates (speeds up like a virus spreading) or decelerates (slows down like a car braking).
- **Curved Line:** When plotted, the graph bends, arcs, or waves; it does not follow a straight diagonal path.
- **Complex Behavior:** These patterns often show explosive growth (exponential) or diminishing returns (plateaus), making them harder to predict than linear trends.
- Patterns can be Exponential, Quadratic, etc.



Real life example of Non-Linear Trends



Do you see anything repeating regularly?



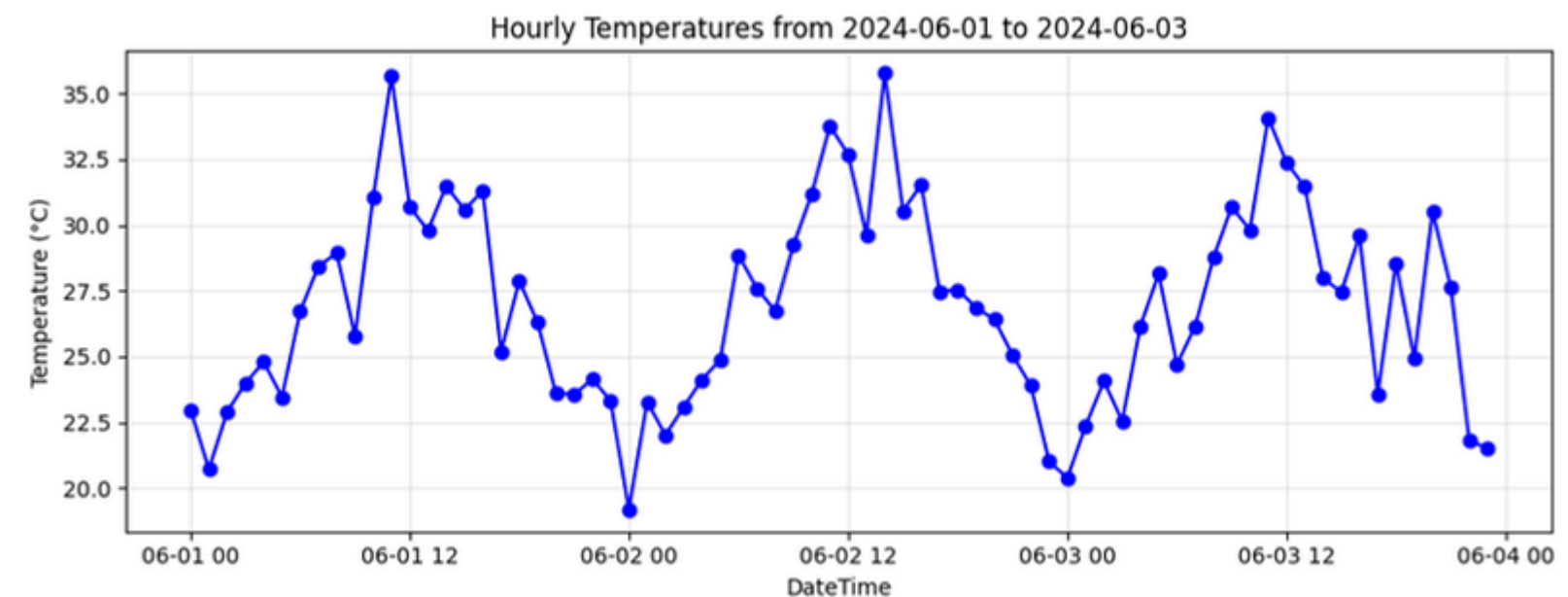
Seasonality

- **Some patterns repeat at fixed intervals:**

- daily
- weekly
- monthly
- yearly

- **Examples:**

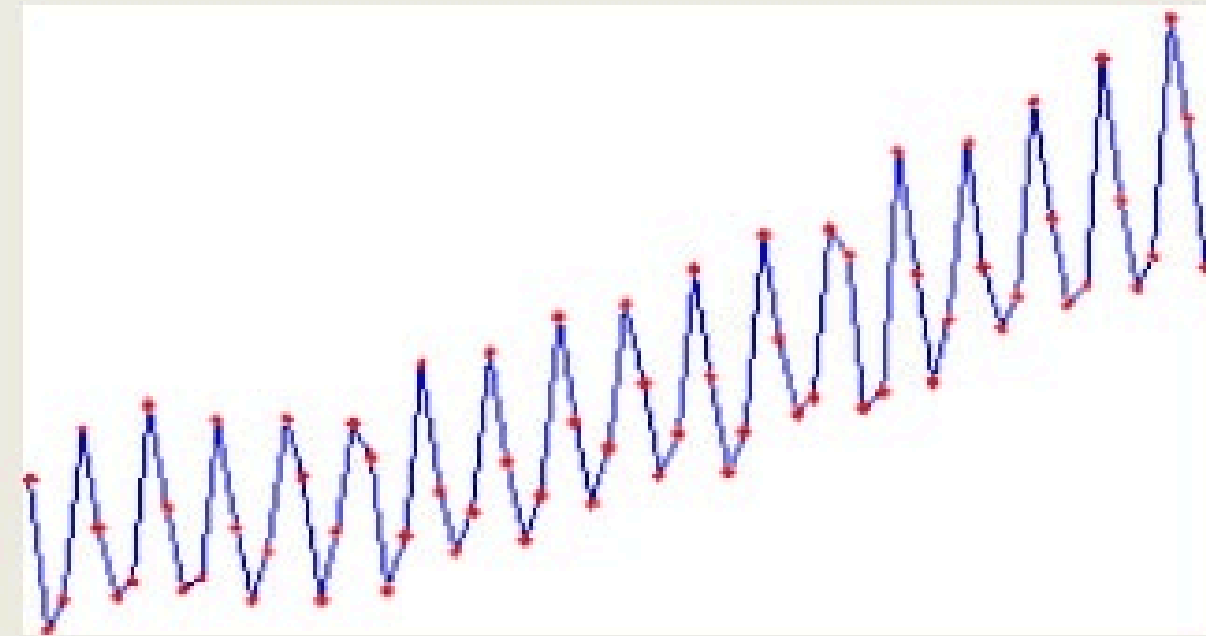
- Weekend dips in traffic
- Festival season spikes
- Summer heat, winter cold



Additive Seasonality

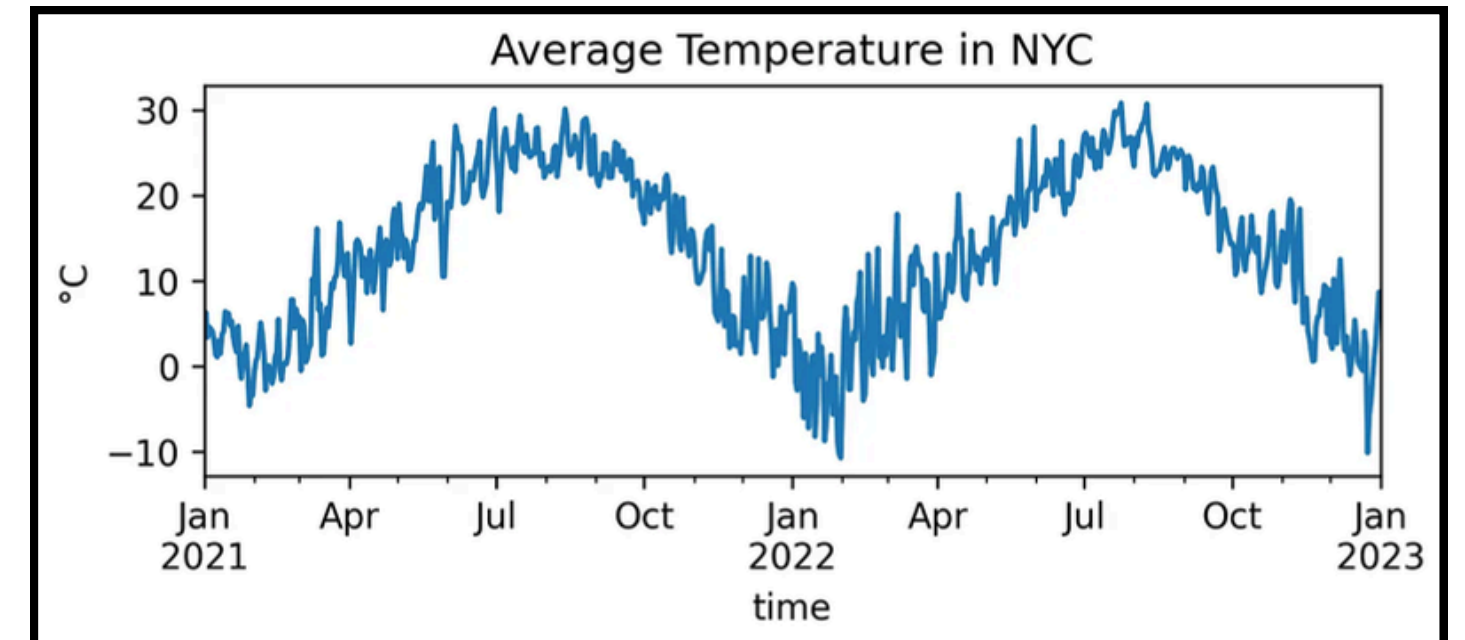
Additive seasonality means that a time series has regular seasonal changes that stay about the same size over time.

Additive



Additive Seasonality

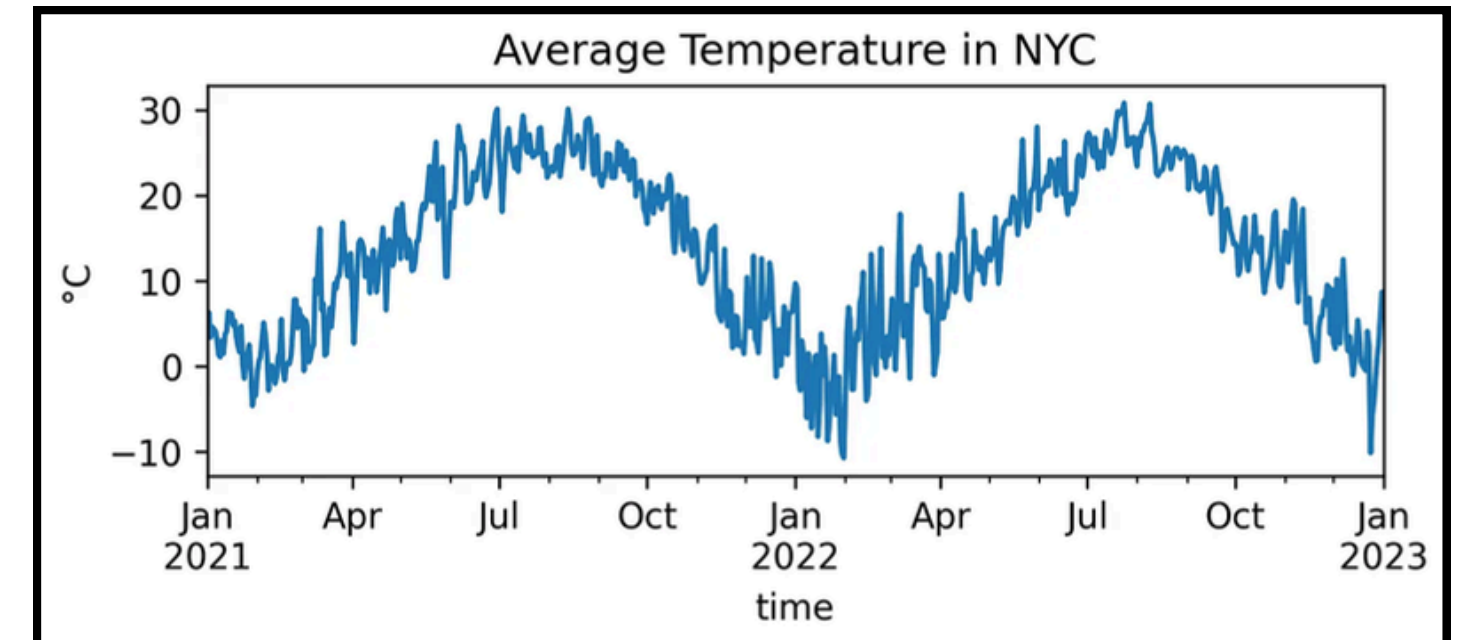
- **Example: Daily Temperature**
 - Summer is always warmer than winter
 - The difference between summer and winter is fairly stable each year
 - Even if the average temperature changes slightly, the seasonal swing remains similar



Additive Seasonality

How It Looks Visually

- Seasonal peaks and troughs have similar height
- Fluctuations do not grow as the trend increases



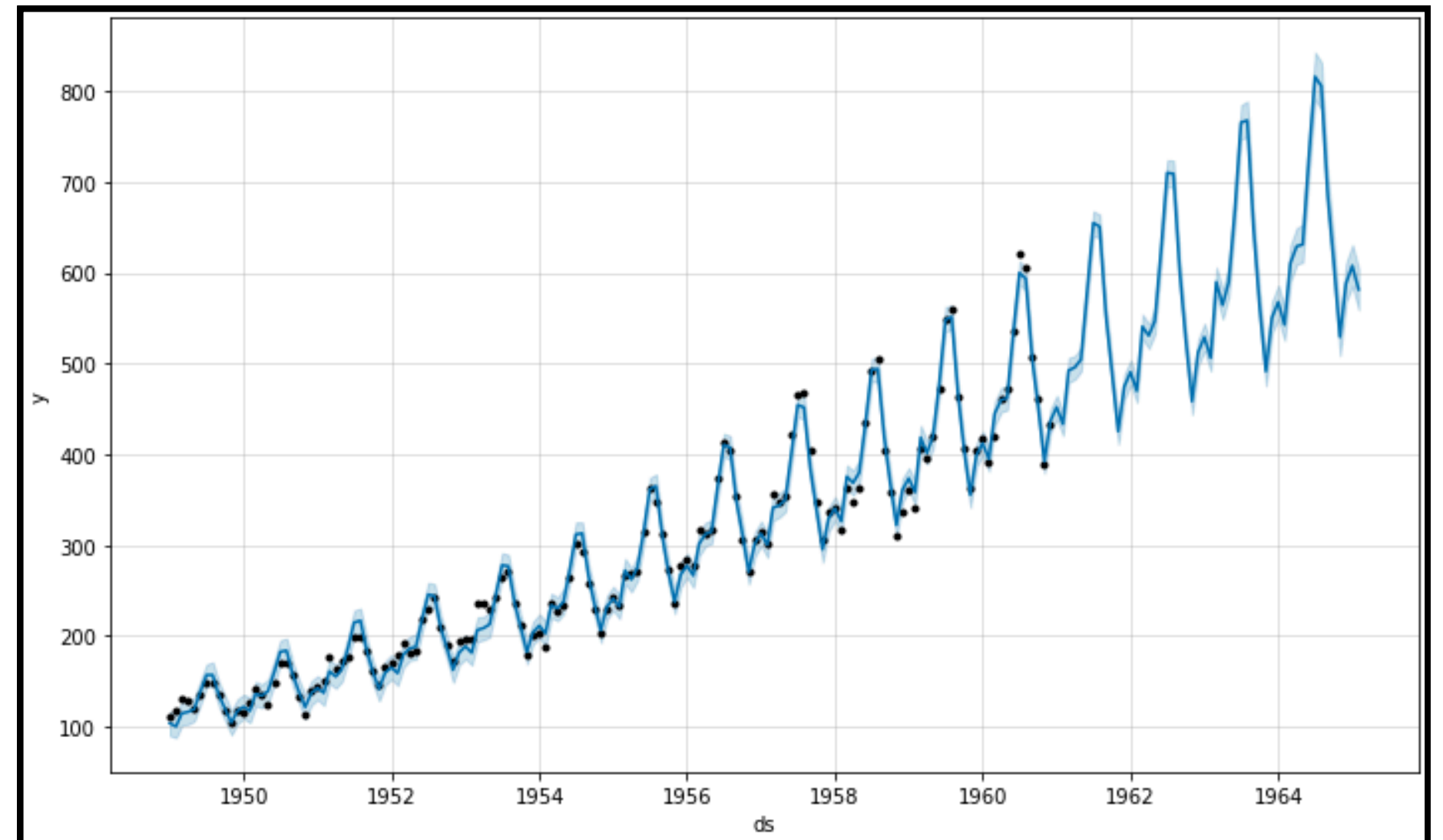
When You Typically See Additive Seasonality

- Temperature data
- Electricity usage in a stable population
- Website traffic with fixed user base
- Manufacturing output with capacity limits

Multiplicative Seasonality

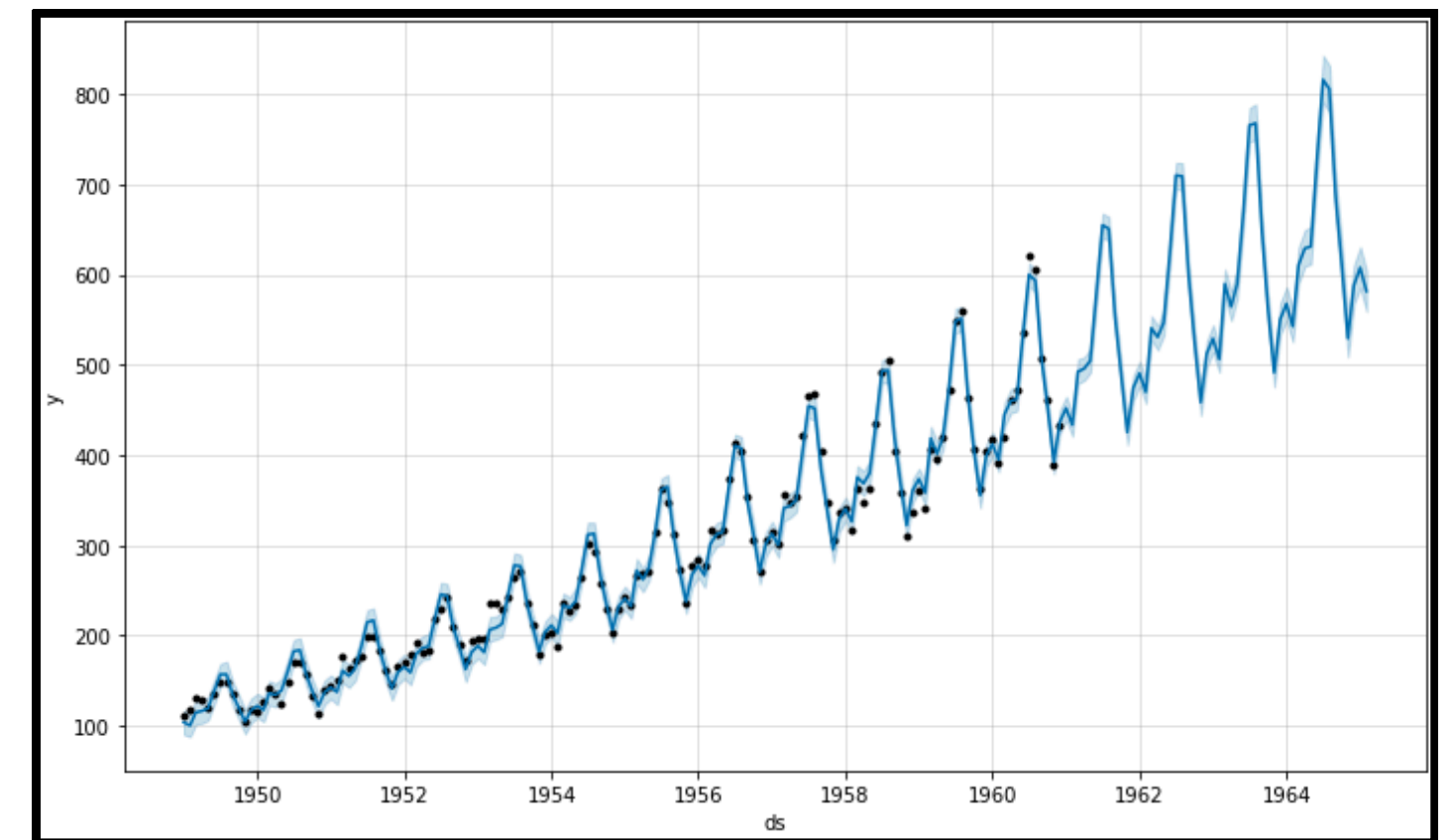
Multiplicative seasonality means that seasonal changes depend on how big the data already is.

The season changes the value by a percentage, not a fixed amount.



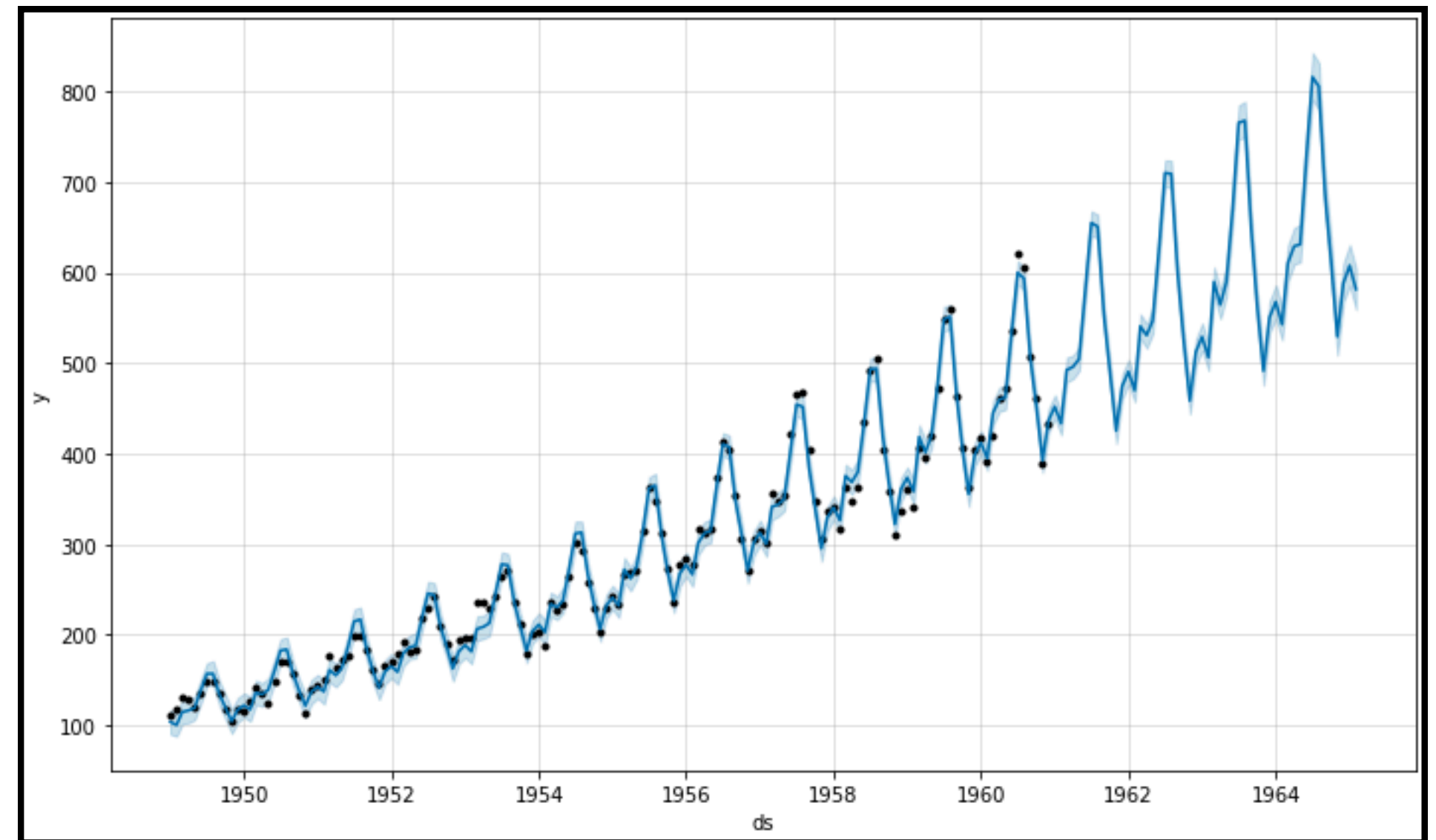
Multiplicative Seasonality

- **Example:** A growing business sells more each year
 - Holiday sales increase in proportion to total sales
 - The seasonal spike becomes larger over time
 - Example:
 - Year 1 December sales: +200 units
 - Year 5 December sales: +2,000 units
 - The seasonality scales with growth.



Multiplicative Seasonality

- **How It Looks Visually**
 - Seasonal fluctuations fan out over time
 - Peaks and troughs grow as the trend increases



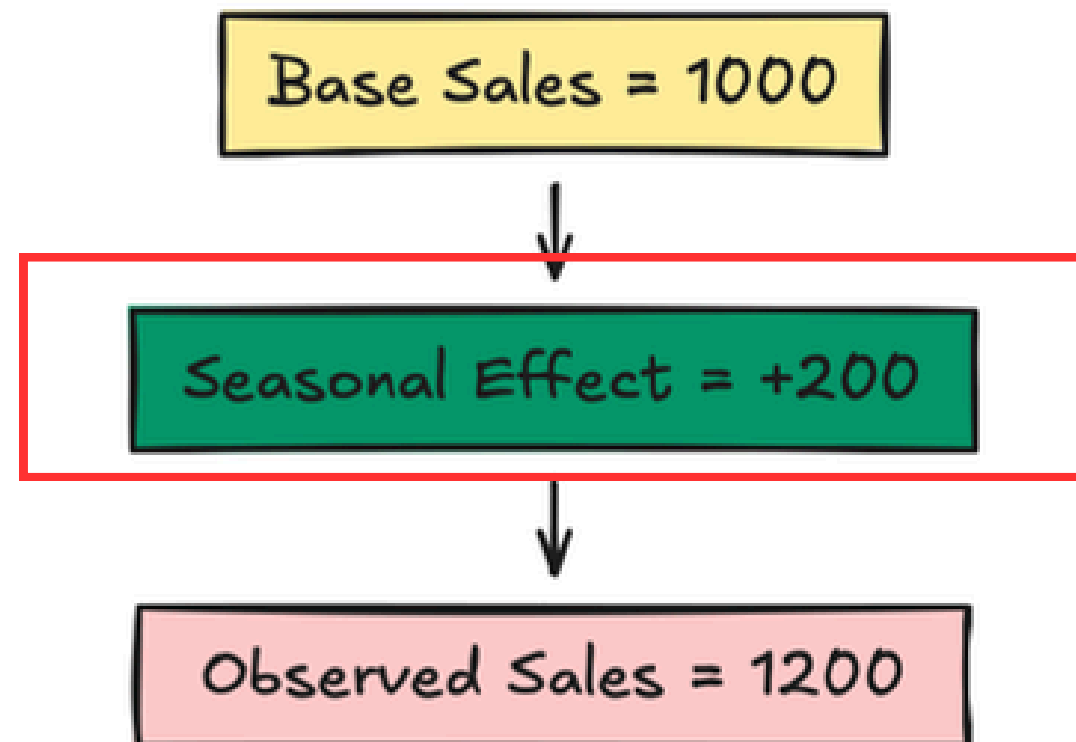
When You Typically See Multiplicative Seasonality

- Retail sales
- Revenue and profits
- E-commerce traffic
- Population-driven demand

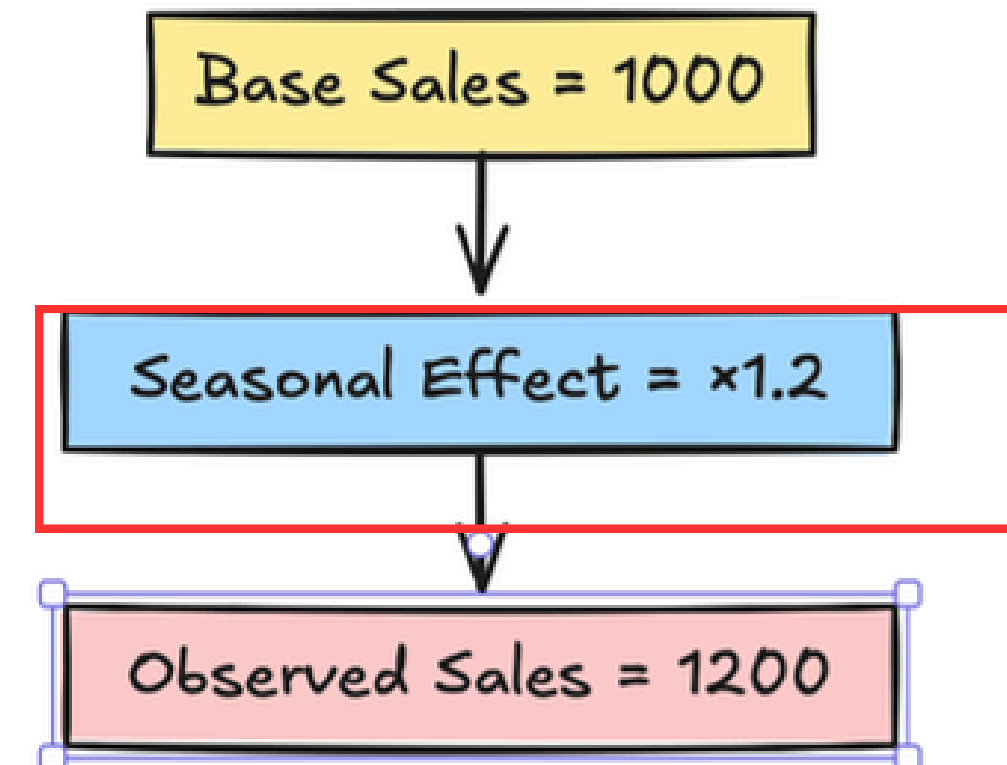
Additive vs Multiplicative Seasonality

Additive Seasonality

Year 1



Multiplicative Seasonality



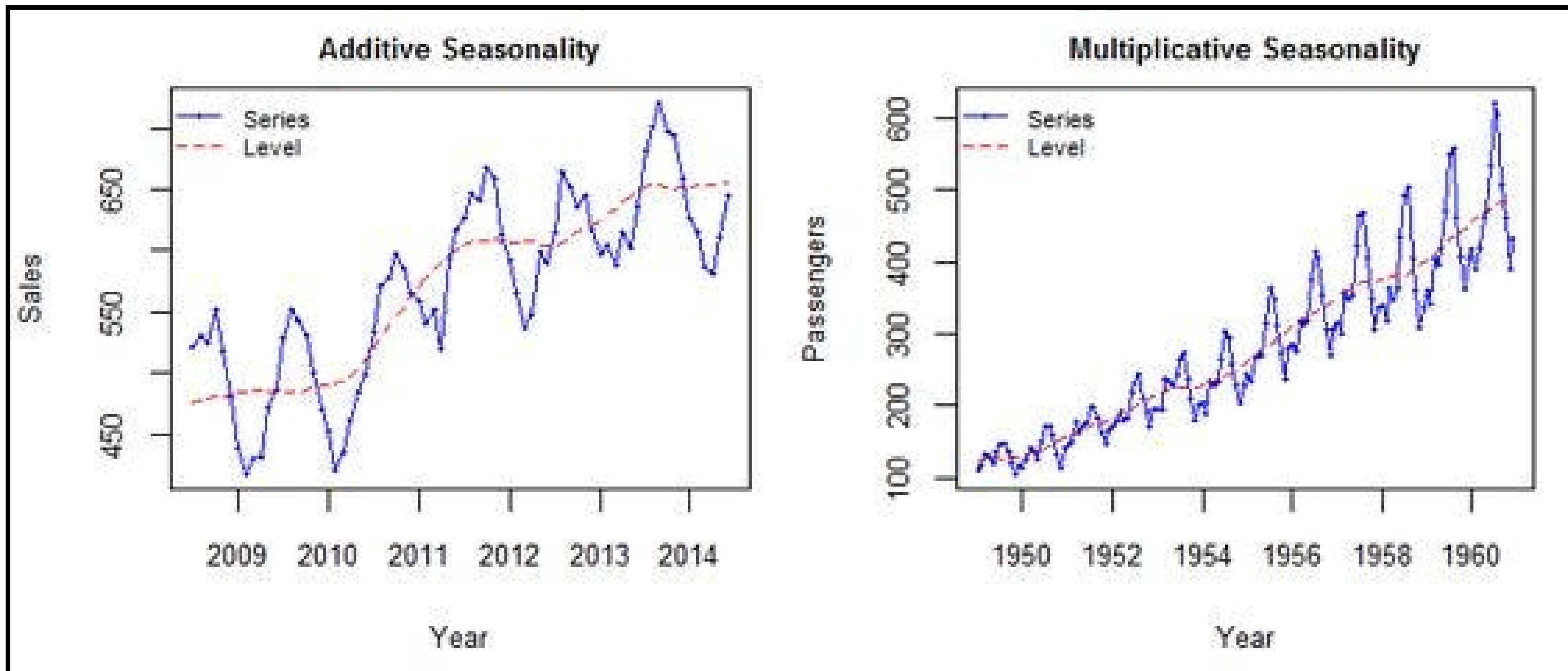
If next year base becomes 2000 — what happens?

- Additive $\rightarrow 2000 + 200 = 2200$
- Multiplicative $\rightarrow 2000 \times 1.2 = 2400$

Same amount $\rightarrow$ Additive

Same percentage $\rightarrow$ Multiplicative

Additive Seasonality vs Multiplicative Seasonality



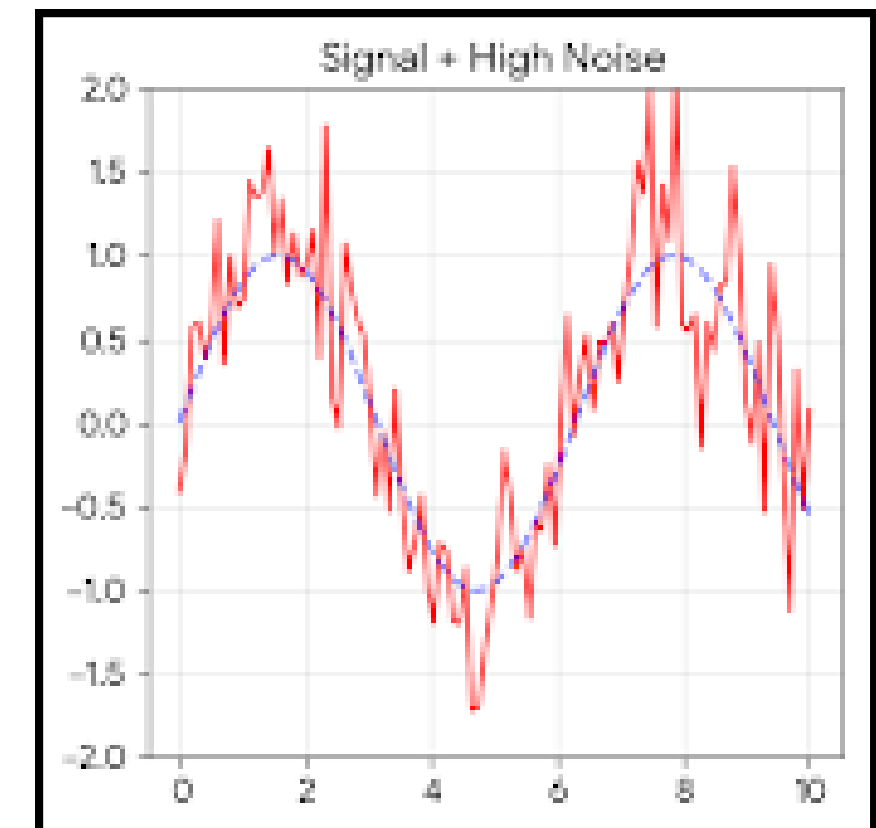
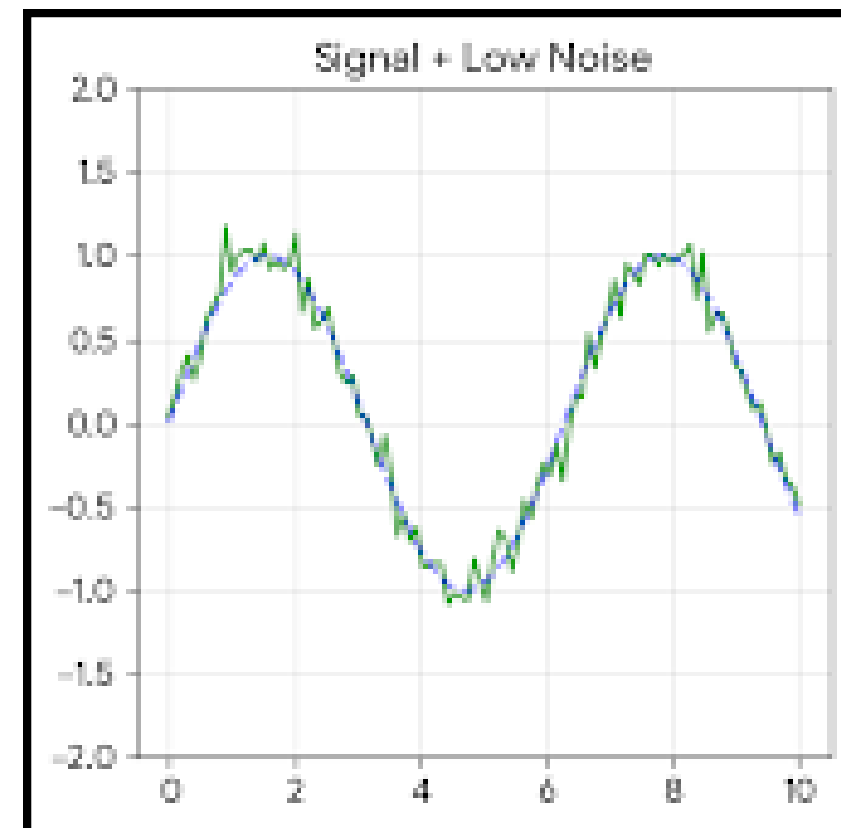
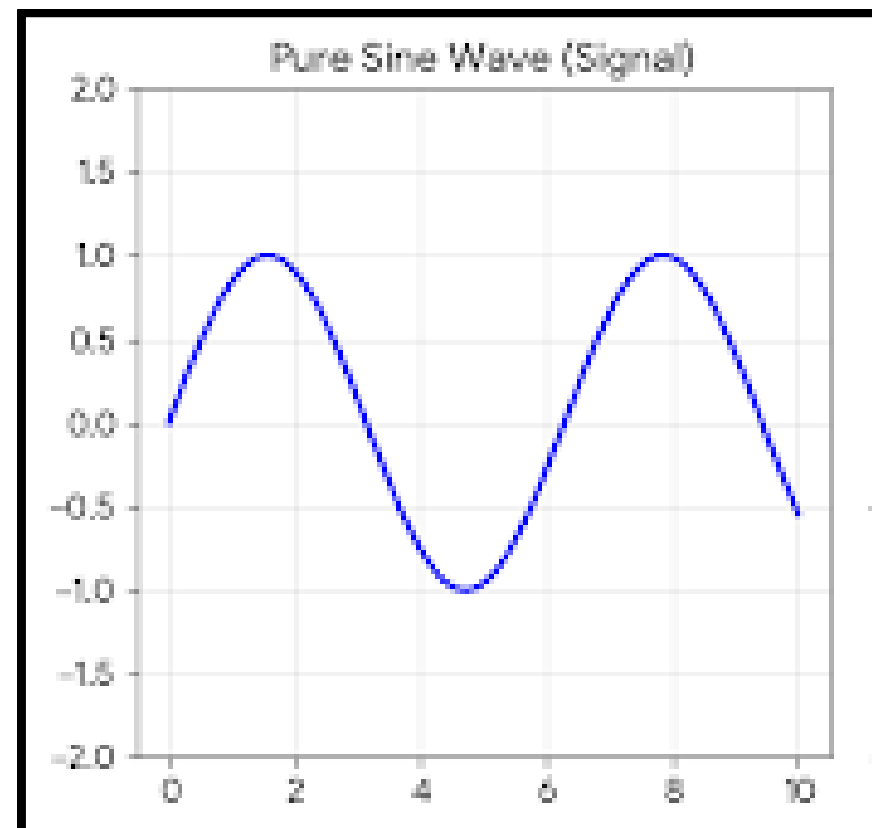
Noise

Noise is the **random, irregular fluctuation** in data that obscures the true underlying pattern and **cannot be predicted**.

- **Unpredictable:** It is the irregular fluctuation in data that has no pattern or specific cause (like a measurement error or a sudden, random event).
- **"Jittery" Look:** On a graph, it appears as erratic spikes, static, or "fuzziness" scattered around the main trend line, making the plot look messy.
- **Masks the Actual Data:** Unlike trends or seasonality, noise contains no useful information; it is the "static" that makes it hard to see the real pattern.

Noise

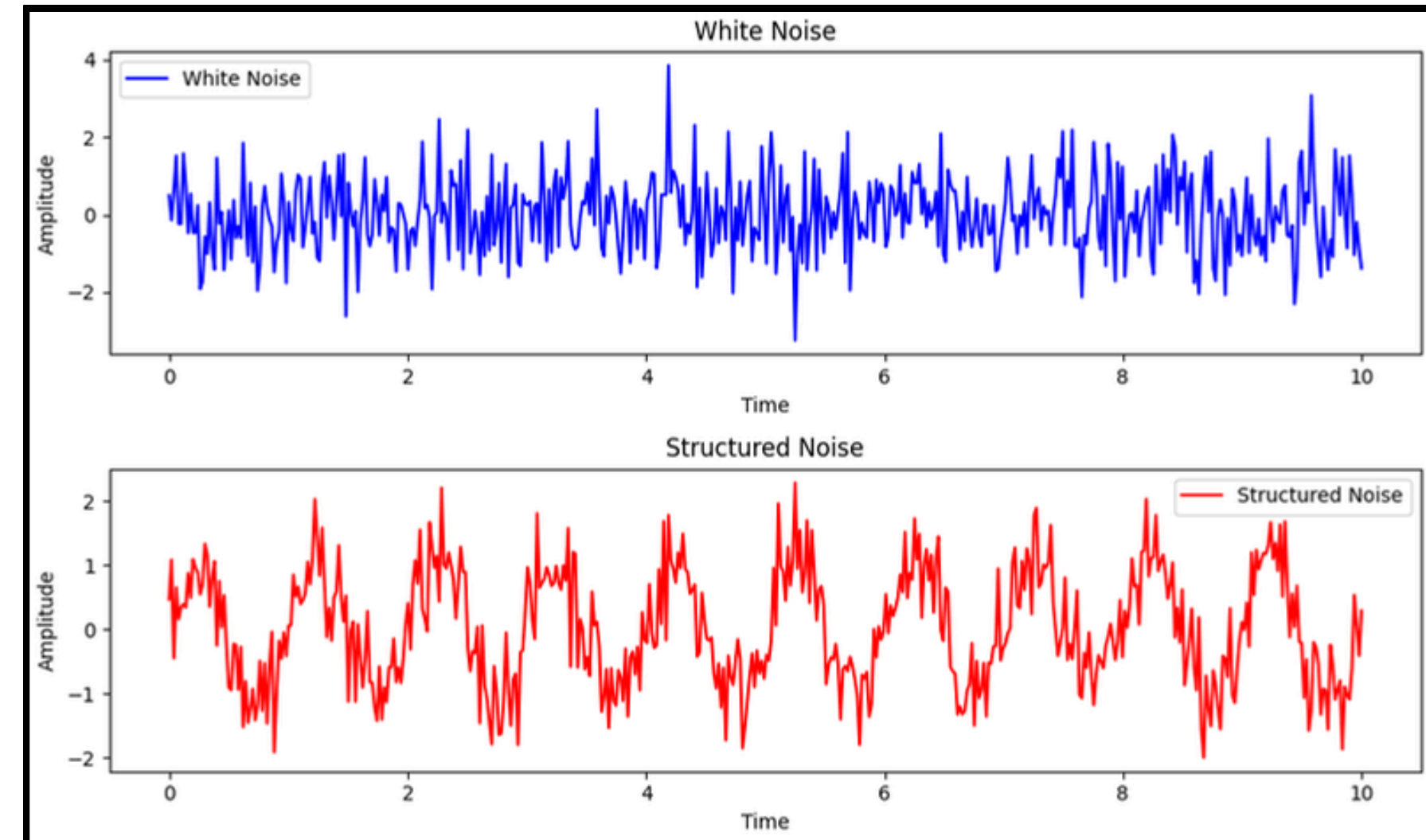
- **Noise includes:**
 - measurement errors
 - unexpected events
 - anomalies
 - randomness



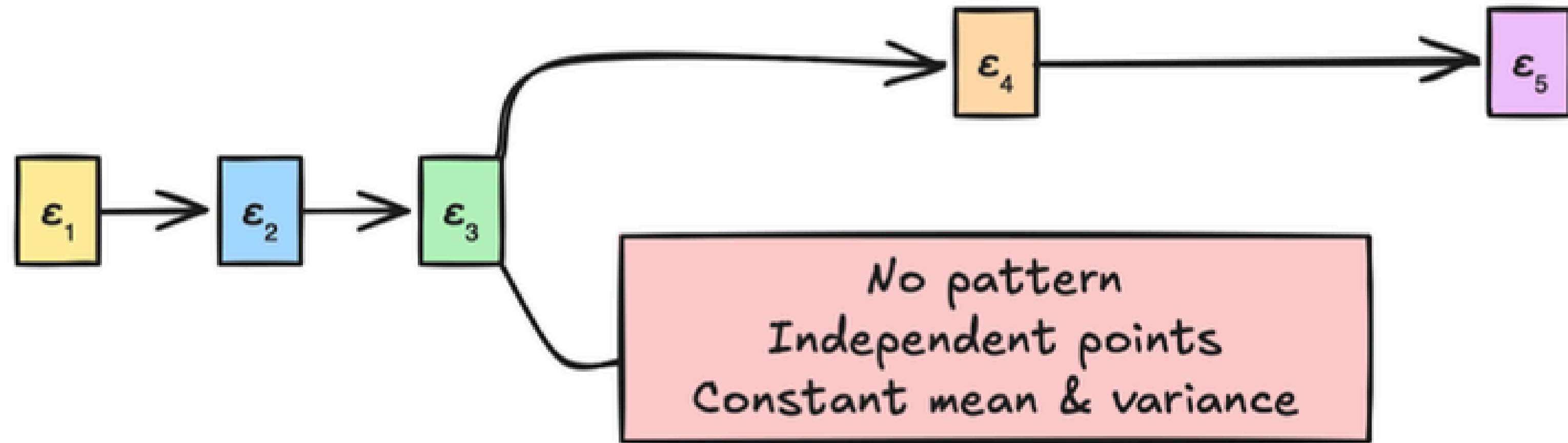
Noise

Two types:

- **White Noise:**
 - Random and unpredictable without any patterns or trends.
 - Each data point is independent of others.
 - Constant mean and variance.
- **Structured Noise:**
 - Shows some pattern or structure (could be cycles, trends, or repeating patterns) is present.
 - Mean and Variance may be variable.

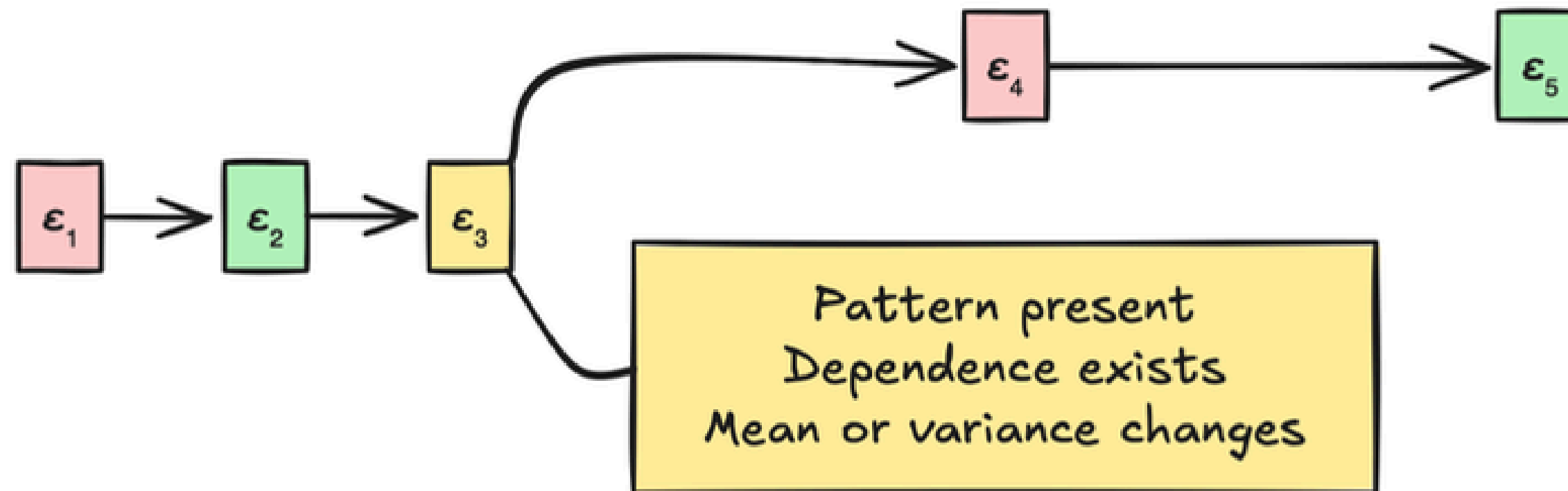


White Noise



Each point is random, unrelated to the previous one, and comes from the same distribution.

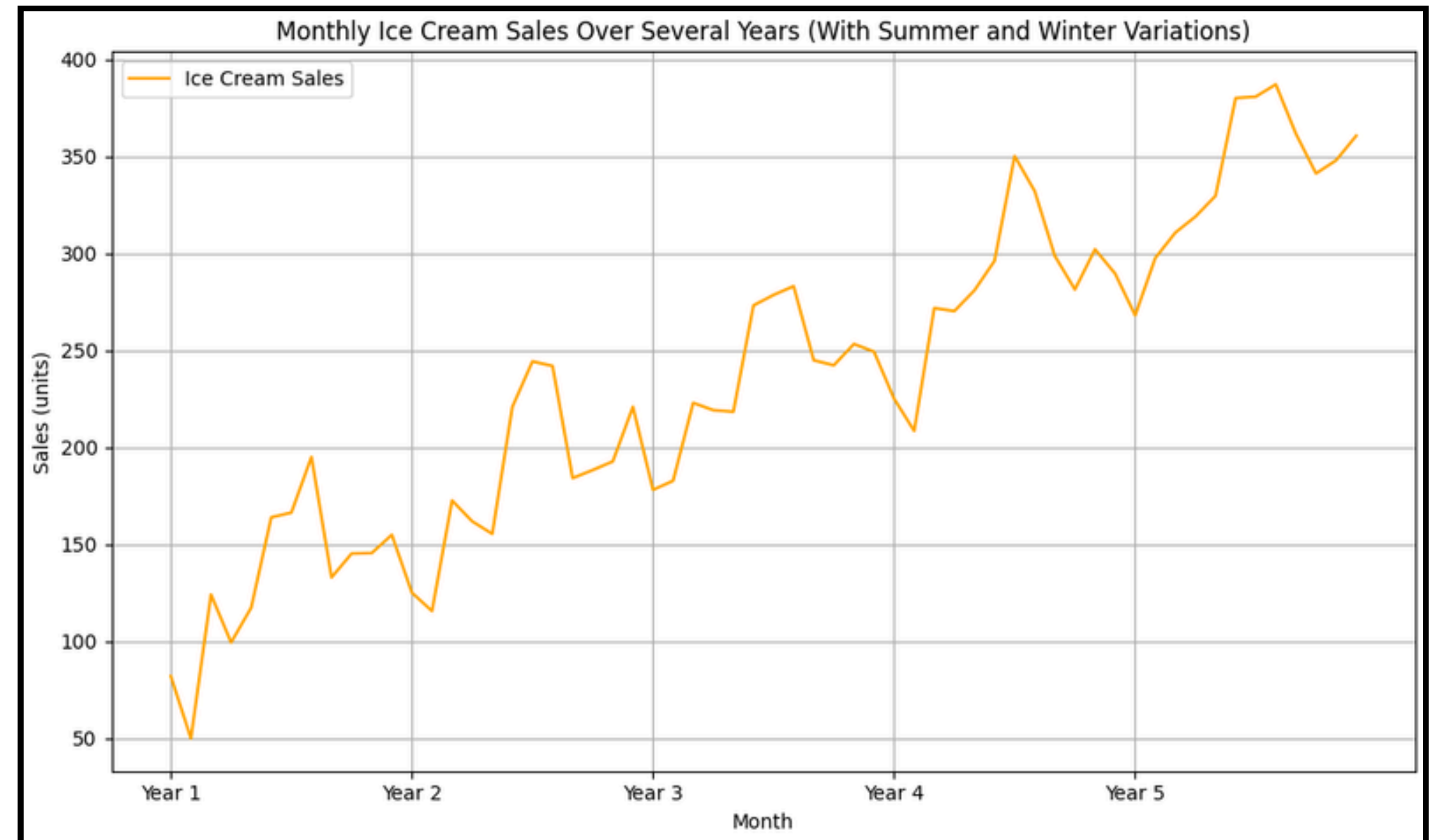
Structured Noise



It looks noisy, but there is hidden structure over time.

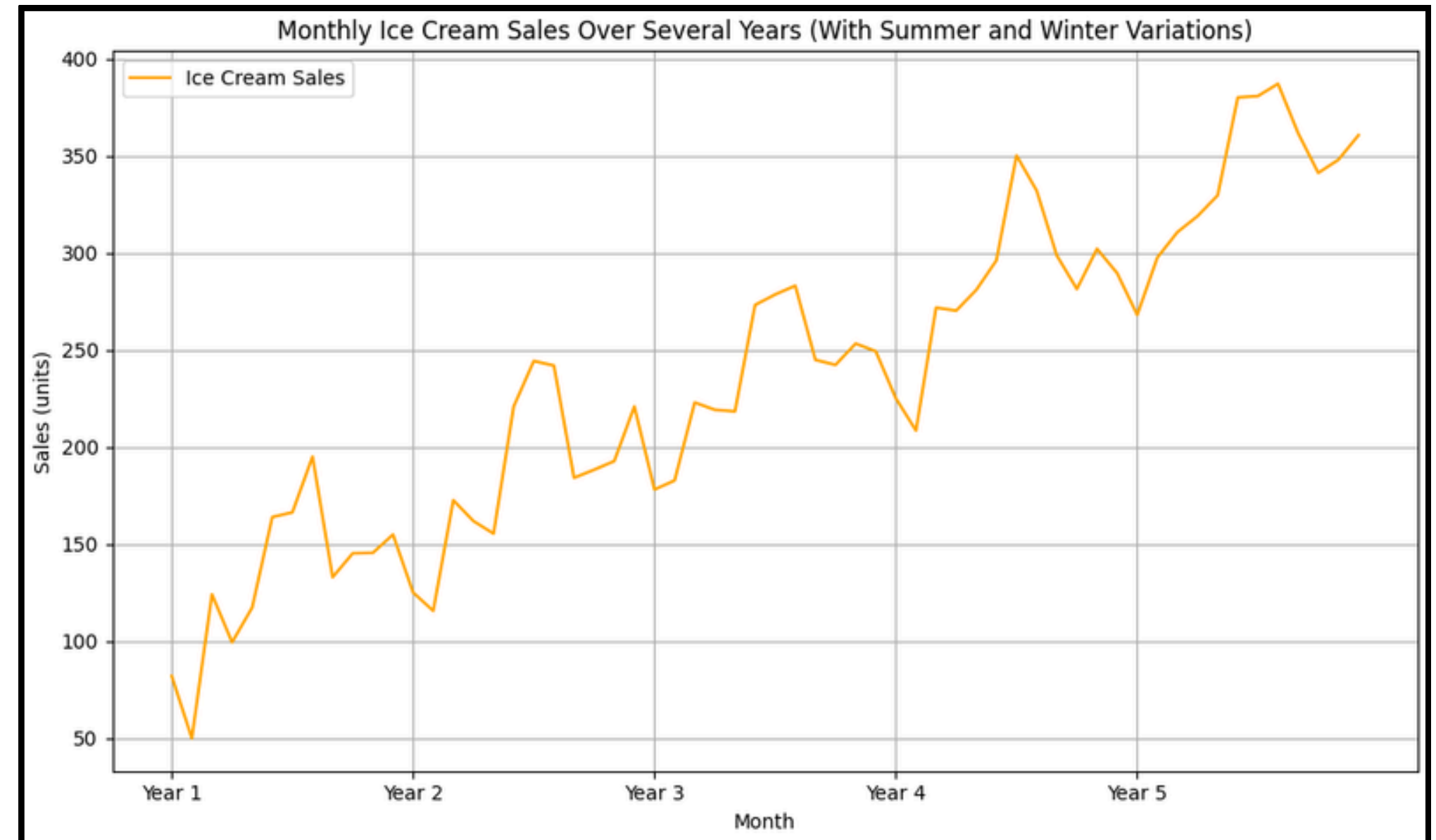
Problem

- You are given a line chart showing **monthly ice cream sales** over several years.
- You notice that sales are generally **increasing, peak every summer, and dip every winter**.
- What **different patterns** do you think are **mixed** in this **single line of data**?



Pattern in the Problems

- Sales seem to be going up over time.
- There's a repeating pattern every year.
- Some months look unusually high or low.



Decomposition

- Time series decomposition is the process of **breaking a time series into simpler components**, each representing a different **type of pattern** in the **data**.
- Instead of analyzing one complicated line, we separate it into:
 - **Trend** – long-term movement
 - **Seasonality** – repeating patterns
 - **Residuals** – leftover randomness

Types of Decomposition

Additive Decomposition

- The time series is modeled as:

$$\text{Time Series} = \text{Trend} + \text{Seasonality} + \text{Noise}$$

- Seasonal effect is added to the trend (a fixed amount).
- Used when seasonal variations are constant and do not change with the overall level of the data.

Types of Decomposition

Multiplicative Decomposition:

- The time series is modeled as:

$$\text{Time Series} = \text{Trend} \times \text{Seasonality} \times \text{Noise}$$

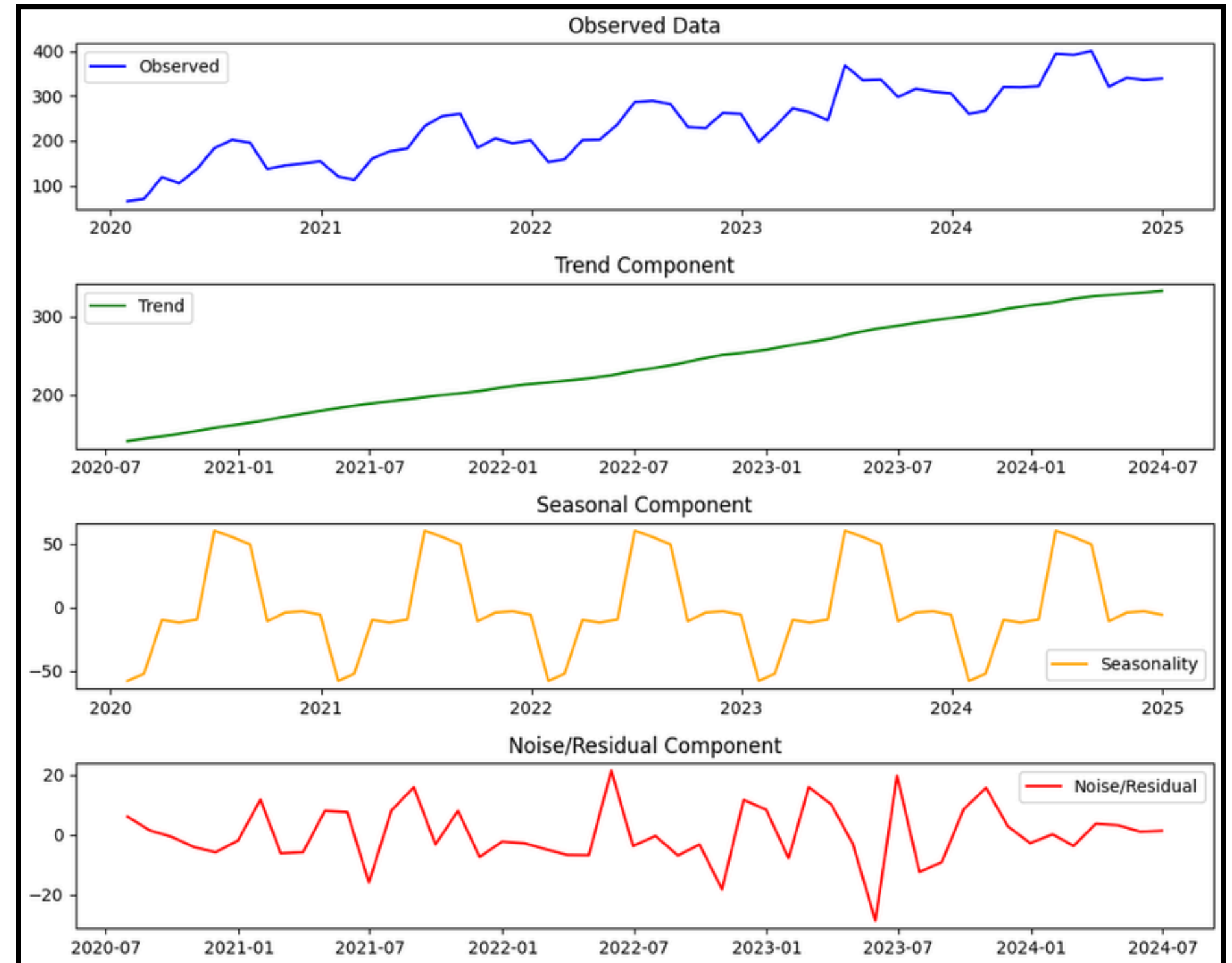
- Seasonal effect is multiplied by the trend.
- Used when seasonal variations are proportional to the trend and increase/decrease as the overall data level changes.

Decomposition Plot

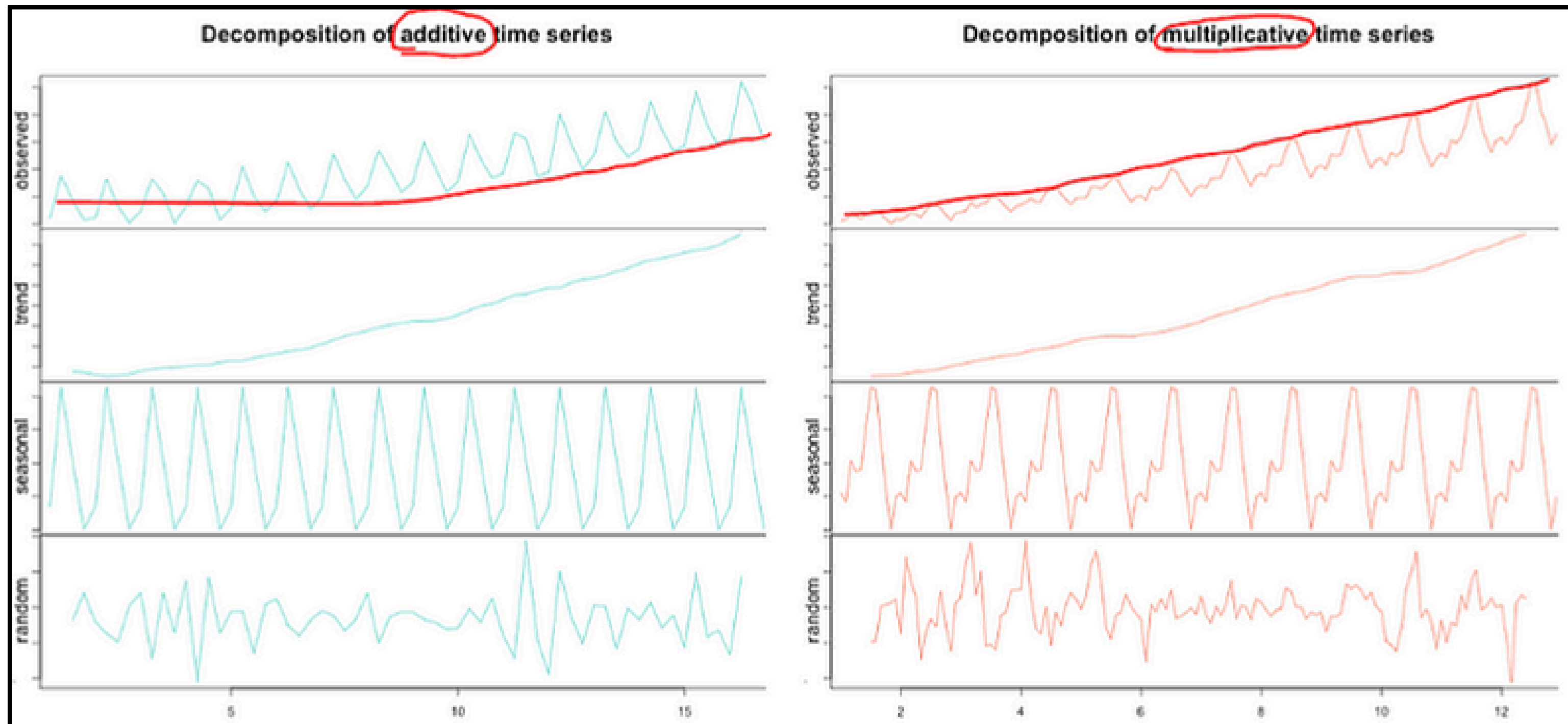
Time series decomposition is usually visualized as a vertical stack of **4** graphs sharing the same time axis.

It visually unzips the complex raw data into simple parts.

- Observed
- Trend
- Seasonality
- Residual/Noise



Additive vs Multiplicative Decomposition



In what other fields are temporal relationships important?

Field where time relation are important

- Stock price prediction.
- Marketing
- Inflation rates over decades
- Music



Marketing

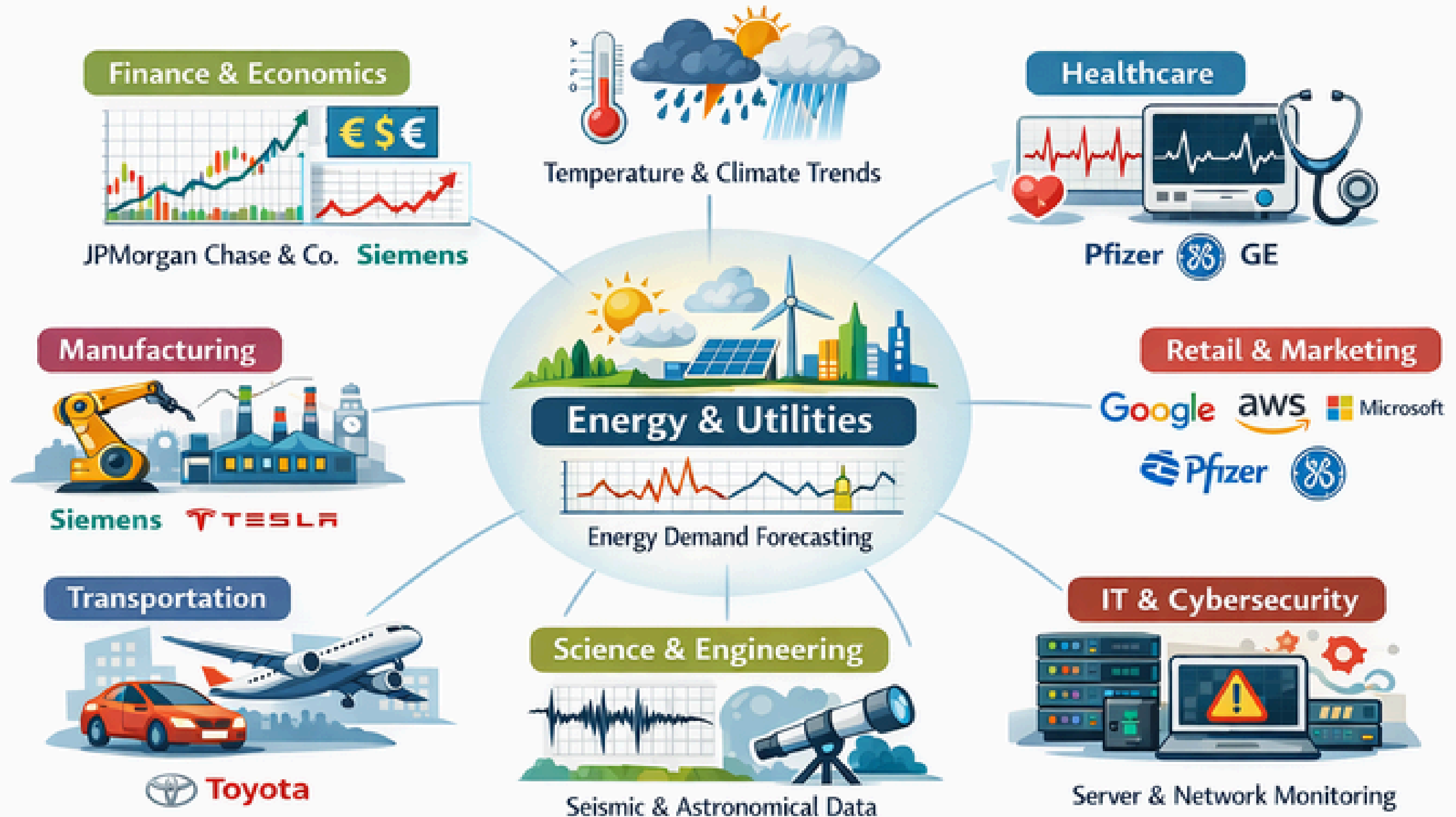


Financial Analysis



Meteorology

Field where time relation are important



Summary

- Time series → Data collected over time, where order matters
- Time index → When each observation occurs (date, hour, timestamp)
- Frequency → How often observations are recorded (hourly, daily, monthly)
- Trend → Long-term direction or movement in the data
- Seasonality → Regular, repeating patterns at fixed intervals
- Noise → Random variation not explained by trend or seasonality
- Decomposition → Separating a time series into trend, seasonality, and noise (often visual)

Time series = signal (trend + seasonality) + noise, indexed by time

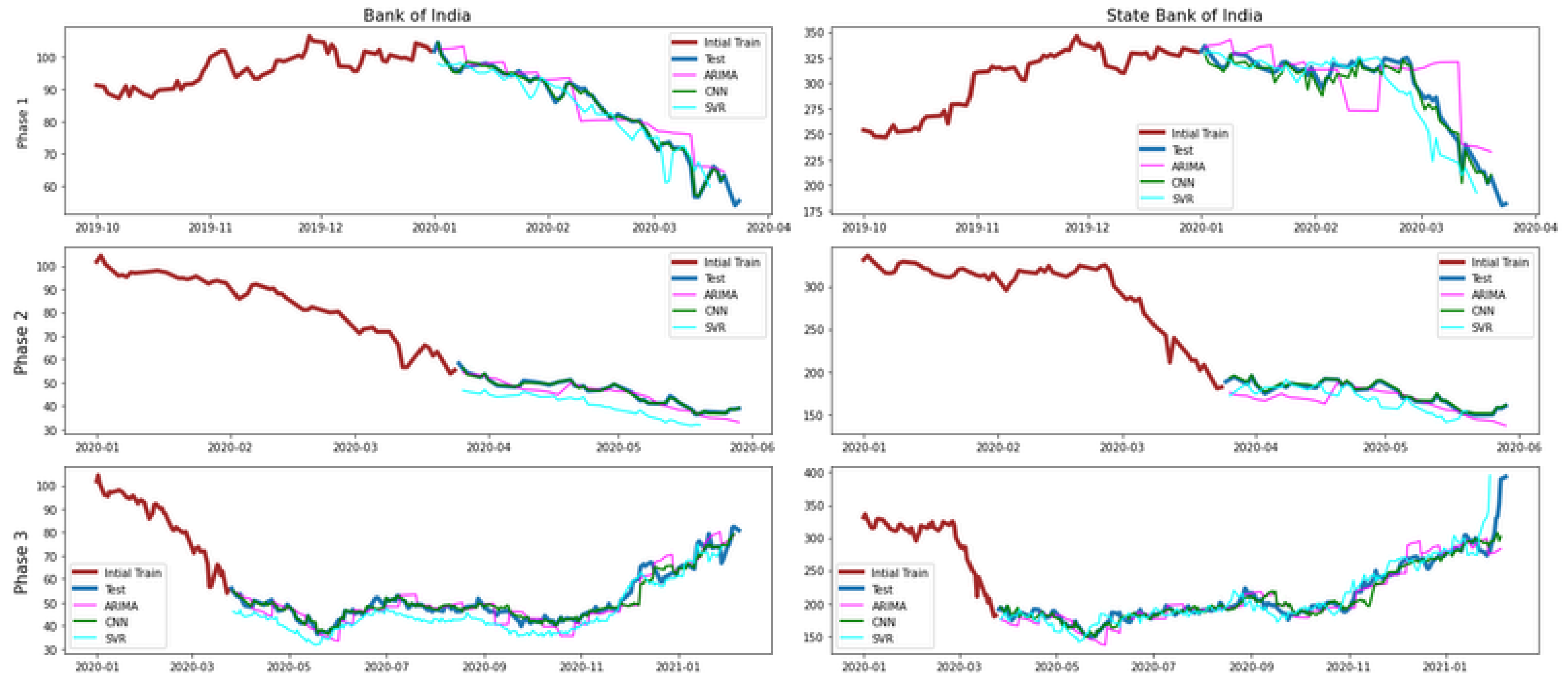
Can we teach an **algorithm** to **predict** a **stock crash** if we **tell** it how **fast** a **virus** is **spreading**?

Actual Volatility vs. COVID-19 Pandemic

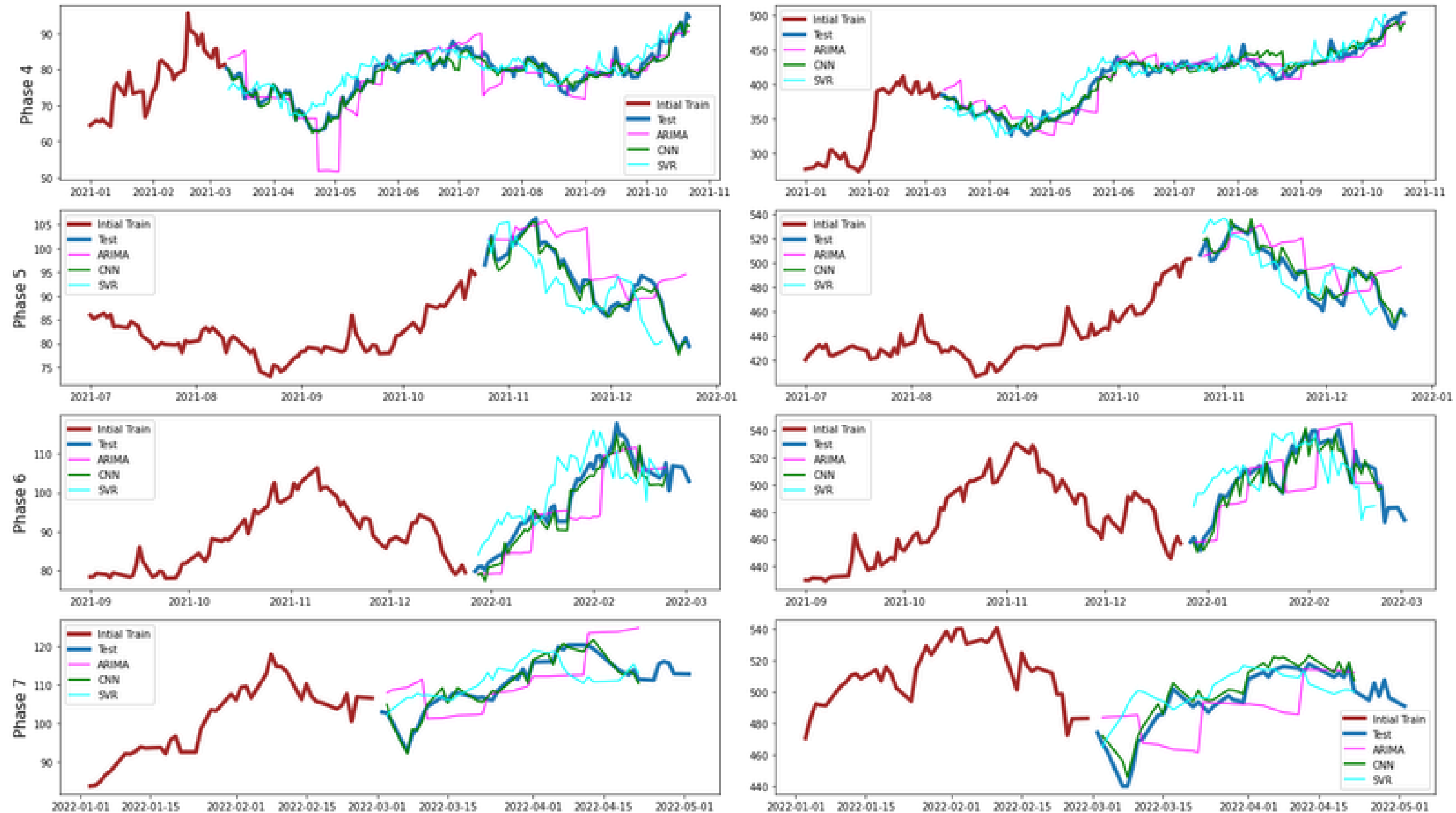
Intervals	P	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
2020-01-02 to 2020-02-12	P1	3.19	40.85	6.85	2.59	7.74	2.20	92.65	1.74	0.50	1.75	9.20	8.74	4.13	13.37	6.51	178.63	45.79	44.02	38.36	19.41	10.92	40.33	14.18	3.00	10.22	82.04	13.43
2020-02-13 to 2020-03-20	P2	15.69	47.95	13.74	17.05	38.85	4.41	170.68	3.95	0.72	16.87	15.15	44.82	9.22	31.23	20.09	584.43	269.43	187.01	187.61	86.52	60.07	125.48	25.49	3.33	18.42	120.19	20.78
2020-03-26 to 2020-05-13	P3	6.31	40.98	4.85	4.41	11.92	0.52	100.35	3.34	0.66	2.58	7.91	8.86	4.27	3.82	5.83	351.02	185.77	153.86	85.05	26.55	33.84	77.69	15.74	3.28	51.21	196.93	78.31
2020-03-14 to 2020-05-29	P4	2.86	9.66	2.20	3.78	23.21	0.48	33.06	1.14	0.16	0.72	2.10	7.46	2.54	4.23	4.27	251.33	74.96	94.47	38.88	18.18	11.19	38.79	4.28	1.14	10.11	33.44	28.05
2020-03-27 to 2020-05-14	P5	6.15	37.11	4.62	4.61	11.08	0.53	97.93	3.23	0.67	2.57	7.66	8.92	3.91	71.75	3.74	351.35	182.52	159.78	86.91	26.41	34.99	71.05	14.42	2.94	48.59	189.36	3.88
2020-05-15 to 2020-06-26	P6	5.80	15.01	4.27	9.30	22.79	3.22	44.47	6.77	0.14	5.36	4.04	12.99	4.54	22.02	4.34	331.90	101.63	125.05	54.10	21.58	22.59	104.06	18.27	7.61	17.27	50.18	11.90
2020-06-29 to 2020-08-07	P7	3.28	17.64	2.22	1.82	8.52	0.68	114.35	2.73	0.31	1.11	5.61	4.76	2.07	30.71	1.68	215.84	75.17	86.50	60.91	87.49	53.29	101.20	24.39	6.89	18.87	173.57	2.30
2020-08-10 to 2020-09-18	P8	3.12	10.68	2.83	7.53	12.68	1.12	45.74	3.79	0.55	7.28	4.88	8.56	1.47	20.66	2.05	209.35	65.05	116.52	79.27	21.38	31.20	61.21	12.41	2.83	12.59	94.86	3.47
2020-09-21 to 2020-11-02	P9	2.06	14.19	1.86	2.82	6.69	0.54	135.87	1.49	0.29	4.95	2.66	7.08	1.13	17.76	1.53	225.86	56.65	149.04	142.13	53.92	25.25	57.21	7.37	2.72	18.21	70.33	2.79
2020-11-03 to 2020-12-15	P10	12.94	28.02	9.41	11.76	16.47	2.46	133.21	4.58	0.29	8.83	8.47	21.02	7.55	17.67	18.27	359.42	144.46	92.20	54.20	26.66	19.11	54.39	13.42	11.05	29.40	191.82	14.03
2020-12-15 to 2021-01-28	P11	7.11	30.97	4.34	3.02	7.32	1.01	101.74	1.39	0.24	7.45	5.71	13.83	6.07	20.48	4.93	263.78	232.92	110.43	178.56	66.06	49.85	30.43	8.37	6.00	16.11	111.69	8.34
2021-01-29 to 2021-02-05	P12	4.33	10.36	4.59	8.44	17.93	0.95	118.42	1.94	0.22	9.62	10.37	17.17	5.14	9.91	6.69	222.65	74.97	181.01	80.18	31.15	27.62	35.28	11.02	6.81	21.94	162.20	62.47
2021-03-10 to 2021-04-27	P13	9.13	8.92	5.37	10.64	44.02	6.93	72.45	8.56	1.19	10.10	11.35	32.85	5.82	7.87	7.30	243.34	172.15	80.03	49.41	30.80	16.83	64.82	18.03	15.17	19.20	98.40	24.95
2021-04-28 to 2021-06-09	P14	5.94	18.30	3.10	6.54	8.31	0.76	231.62	3.94	0.75	12.07	13.96	5.42	2.18	2.79	1.52	201.70	57.40	42.16	57.91	48.37	13.03	24.00	10.03	6.84	7.12	176.47	9.28
2021-06-10 to 2021-07-22	P15	3.07	11.81	2.14	3.52	14.08	1.00	399.39	3.57	2.98	8.23	9.11	10.99	2.83	4.36	3.17	175.25	58.42	53.64	212.21	48.77	66.46	49.28	29.21	16.74	36.23	106.97	19.91
2021-07-23 to 2021-09-03	P16	5.83	24.29	15.77	26.20	43.82	1.81	243.41	1.33	20.78	4.28	9.07	18.66	4.07	12.51	2.99	305.69	77.42	44.46	85.45	26.76	44.75	23.71	9.37	6.02	27.65	148.79	23.84
2021-09-06 to 2021-10-19	P17	2.52	12.70	1.70	6.19	9.47	1.02	112.27	1.15	6.52	8.00	3.26	7.02	2.72	4.01	0.56	105.10	44.51	59.26	63.36	46.04	13.16	10.13	4.99	2.99	6.44	59.23	6.32
2021-10-20 to 2021-10-22	P18	7.72	11.54	4.02	11.94	26.07	1.71	584.11	3.31	80.66	14.98	11.21	21.05	6.11	13.80	4.11	313.28	202.89	100.10	56.56	33.42	6.11	41.27	15.30	7.64	24.62	177.91	21.28
2021-10-26 to 2021-12-08	P19	2.74	6.75	5.04	9.90	11.80	1.70	229.87	2.88	35.07	11.66	9.51	17.65	5.93	10.46	2.11	86.77	89.81	67.43	30.51	38.81	5.93	52.69	3.35	8.29	13.42	77.65	11.83
2021-12-09 to 2021-12-23	P20	6.63	33.31	11.33	14.12	17.68	7.74	233.53	2.25	17.68	13.22	9.69	26.71	9.85	21.94	2.55	454.45	119.85	118.39	91.23	77.26	83.07	21.20	7.99	10.64	27.19	233.11	21.94
2021-12-28 to 2022-02-08	P21	2.65	11.10	2.55	5.07	8.47	2.18	79.13	3.29	27.30	19.56	12.21	14.96	4.77	11.68	1.48	144.50	48.17	46.62	81.98	21.16	12.07	33.71	5.89	9.10	17.92	81.40	11.68
2022-02-09 to 2022-02-23	P22	7.75	16.66	5.37	21.53	41.71	5.07	127.68	2.54	19.29	38.23	4.82	22.06	7.61	12.93	1.76	273.50	181.72	62.49	78.44	62.72	22.90	28.38	20.00	12.83	34.95	138.73	38.25
2022-03-03 to 2022-04-22	P23	2.58	7.34	1.30	3.42	9.23	1.32	126.39	1.51	6.86	3.96	2.69	5.72	1.02	1.98	0.39	200.55	35.38	20.65	63.77	24.98	13.25	32.05	1.31	5.66	9.02	49.60	22.39

- Banking sector experienced moderate levels of volatility during all events
- Each sector and stock tend to fall during the second phase of every crisis but begin to move upwards during the last phase

Actual Volatility vs. COVID-19 Pandemic



Actual Volatility vs. COVID-19 Pandemic



Please fill the feedback form.

Thank You